Shared SDG Labels Do Not Imply Shared Framing — Embedding-Based Register-Topic Decomposition of Research-Policy Divergence Across the SDGs

Abstract

Research and policy can address the same Sustainable Development Goal while framing it in different ways. Existing assessments of research–policy alignment usually measure whether both sides mention the same goals, but this does not show whether they discuss those goals similarly. This paper asks how academic AI-for-sustainability research and policy discourse differ in both the SDGs they prioritise and the way they discuss each SDG.

The study compares a large corpus of research abstracts with policy documents covering all 17 SDGs. It measures each corpus's distribution of attention across the goals and compares research and policy language within each SDG. The analysis separates differences associated with substantive topics from differences associated with corpus-specific writing conventions. The framework uses supervised text classification and embeddings, with checks across alternative encoders, classifiers, retrieval strategies, sampling levels, and policy-source collections.

The results show different coverage profiles. Research concentrates especially on SDGs 3, 4, 9, and 16, whereas policy discourse places greater emphasis on SDGs 16, 17, and 13. Differences in coverage alone do not explain the raw semantic differences between the two corpora. After adjustment for corpus-associated linguistic variation, the relationship between coverage and semantic divergence becomes more apparent. The adjustment also changes the ranking of some goals, most notably SDG 17, showing that writing conventions can either inflate or conceal substantive differences.

These findings show that shared SDG labels are insufficient evidence of research–policy alignment. Alignment should be assessed along at least two dimensions: which goals each community prioritises and how each community frames those goals. The framework provides a reproducible way to identify where apparent semantic disagreement reflects substantive differences, corpus-associated language, or both.

Keywords: research-policy alignment; Sustainable Development Goals; semantic similarity; transformer embeddings; measurement protocol; bibliometrics.

Data and code availability: The analysis code and the frozen, processed outputs are version-controlled at `https://github.com/qmanhbeo/msc-dissertation-leo`.

Technical Summary

Shared SDG labels do not mean research and policy *frame* a goal similarly. This paper contributes a measurement framework that separates research-policy divergence into two dimensions: coverage (which SDGs each corpus prioritises) and within-SDG semantic framing (how differently the same SDG is discussed). Framing is composed of topic (substantive content) and register (communicative conventions).

The paper embeds 62,173 labelled reference texts, 3,105,144 research segments, and 40,597 policy segments with a frozen Sentence-BERT encoder. Labelled segments are used to train a single-label logistic regression classifier to score research and policy text segments. The study then trains an SDG-stratified research-policy LR classifier to subtract the linear directions that separate research and policy from the original semantic space using Iterative Nullspace Projection (Ravfogel et al., 2020). What remains is called the adjusted semantic space, interpreted as containing the substantive topic. Texts are re-scored in this adjusted space. The difference between the raw and adjusted embeddings is called the register component.

For each SDG in each corpus, coverage is its share within the corpus and semantic centroid is the mean value of the embeddings labelled that SDG. For each SDG, coverage gap is the coverage difference between the two corpora and raw semantic gap is the cosine distance between research and policy centroids. Adjusted semantic gaps are computed analogously on the projected embeddings, and the register component is the difference between the raw and adjusted gaps. Correlation tests between H1a) absolute coverage gap, H1b) signed dominance, H1c) research coverage, H1d) policy coverage and within-SDG semantic gaps (raw, adjusted, register) are calculated with two-sided Monte Carlo permutation p-values (100,000 resamples).

The SDG classifier achieves macro-F1 = 0.816 on a held-out test set and reproduces the high research coverage of SDGs 3, 4, and 16 reported in prior inventories; SDG 9's second place is plausibly an artefact of the AI-focused retrieval. The policy-side ranking (top SDGs: 16, 17, and 13) is reported here. The raw coverage-semantic correlation is null ($\rho = -0.012$, p = 0.959). However, the adjusted semantic gap is positively associated with coverage ($\rho = 0.544$, p = 0.025), whereas the register component has a negative, non-significant association ($\rho = -0.390$, p = 0.120). Policy coverage is the strongest directional predictor of the adjusted semantic gap ($\rho = +0.589$, p = 0.015), while research coverage shows no consistent signal ($\rho = +0.267$, p = 0.295).

These results are stable across encoders, classifier family, retrieval strategy, segment cap, and policy-source family (source families agree on the raw-gap pattern; Section 4.4), apart from one sign flip in the concept-retrieved track (Section 5.5). To conclude, coverage and framing should be measured and reported as separate axes of divergence. Shared labels are insufficient evidence of shared framing.

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1 INTRODUCTION

Research-policy alignment is often inferred from whether the two sides address the same topics (Bornmann, Haunschild, and Marx, 2016), but shared coverage does not mean comparable framing. Research and policy rarely co-occur on a topic: Only 1.2% of 191,276 climate-change publications carry at least one policy mention (Bornmann, Haunschild, and Marx, 2016), and shared terminology conceals differences in term salience and focal concerns (Toney-Wails, Curlee, and Probasco, 2024). A paper and a policy document can both be labelled the same SDG while emphasising different problems, actors, and interventions. Legislative agendas, ESG disclosures, and funding decisions increasingly treat SDG tags as evidence of alignment (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022) while this assumption is rarely tested directly.

Existing methods for mapping research to the Sustainable Development Goals (SDGs) operate mainly at the level of lexical or bibliometric correspondence: keyword dictionaries, Boolean queries, or citation and mention counts (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024). Broader AI-SDG assessments compound this, resting on expert opinion or publication counts rather than on the texts themselves (Vinuesa et al., 2020). They identify whether a text addresses an SDG and leave its framing unexamined.

Coverage and framing gaps appear beyond AI: conservation science shows 59% of policy-prioritised topics in fewer than 15 papers (McCarthy et al., 2026), corporate AI research concentrates on pre-deployment problems (Strauss et al., 2025), and global AI metrics emphasise technical and economic performance where policy stresses ethical and social dimensions (Sioumalas-Christodoulou and Tympas, 2025).

This paper proposes that research-policy alignment has at least two separable dimensions: coverage divergence (which SDGs each corpus prioritises) and semantic framing divergence (how differently they discuss the same SDG). Applied to the AI-SDG scientific and policy literature, this framework advances such mapping from Level 1 (lexical correspondence) to Level 2 (observed textual-semantic proximity in embedding space, Section 2.3). A novel SDG-stratified INLP adaptation separates register from topic.

The central research question is: To what extent do academic AI-for-sustainability research and policy discourse differ in SDG coverage and semantic framing across the 17 SDGs, and how are these two dimensions related?

2 LITERATURE REVIEW

2.1 Previous AI-SDG Bibliometrics

SDG scientometrics and AI-specific inventories (Vinuesa et al., 2020) converge on one robust finding: SDG 3 (Good Health) and SDG 7 (Affordable and Clean Energy) lead, followed by SDGs 4, 13, 11, and 16. The evidence base includes a citation-based corpus of roughly 25,000 MDG and SDG publications (Bautista-Puig et al., 2021) and 792 AI-and-SDG articles (Gohr et al., 2025). None of these studies measures within-SDG semantic framing.

SDG publication mapping depends on method. Armitage, Lorenz, and Mikki (2020) shows that what counts as SDG-relevant depends on interpretive choices about queries, labels, and databases. Kashnitsky et al. (2024) find no single best approach and no single "golden" validation set across Boolean, machine-learning, and hybrid systems. Keyword baselines still omit publications that never state an SDG keyword explicitly, keeping the corpus boundary query-dependent (Yin et al., 2025). Embedding-based retrieval reduces some vocabulary brittleness but stays conditional on corpus construction, centroids, and model choice.

The SDGs are interdependent by design, yet no operational framework specifies how to manage these interdependencies (Nilsson, Griggs, and Visbeck, 2016). SDG 17 is the partial exception. Le Blanc (2015) treats SDG 17 (Partnerships for the Goals) as the dedicated means-of-implementation goal (finance, trade, technology transfer, capacity building) and excludes it from cross-goal mapping. Its discourse concerns coordinating and resourcing implementation itself, a conceptual space AI research rarely addresses, unlike every other SDG whose discourse centres on a shared substantive topic. Single-SDG assignment is therefore a simplifying assumption.

2.2 Research-Policy Alignment in AI and Sustainability

A growing literature documents structural tensions between what AI research produces and what sustainability policy requires. Corporate AI research concentrates on pre-deployment technical problems, creating a gap between research priorities and the deployment impacts governance must address (Strauss et al., 2025). Research and policy communities can share trustworthy-AI terminology while differing in term salience and focal concerns (Toney-Wails, Curlee, and Probasco, 2024).

Scientometric work counting policy-document mentions gauges connection without examining framing (Bornmann, Haunschild, and Marx, 2016).

Theory explains why such divergence is the default rather than a failure. Knowledge systems support action when they produce information that is simultaneously salient, credible, and legitimate (Cash et al., 2003). Because these are properties of how information is framed and packaged rather than of which topics it covers, they motivate measuring framing alongside coverage instead of only topical presence. The epistemic-communities framework explains why shared framing matters for policy coordination under uncertainty: expert communities influence policy by articulating cause effect relationships and framing issues for collective debate (Haas, 1992). Haas's concept is broader than semantic proximity, involving shared normative and causal beliefs, validity criteria, and a common policy enterprise. The present study therefore measures only the weaker condition of observed semantic proximity, not an operationalised epistemic community.

The benchmark itself is not neutral. The SDG framework is a politically negotiated instrument with documented limitations: goals can create incentives to over-focus on target achievement (Fukuda-Parr, 2016), summit documents remain state-centric and avoid causal responsibility (Bexell and Jönsson, 2017). Reported SDG engagement is frequently superficial, so reported alignment should not be conflated with substantive commitment (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022). If research follows Mode 1 knowledge production (problems set within academic disciplines) while Mode 2 answers problems arising in application contexts (Gibbons et al., 1994), divergence can mark productive exploration before policy absorbs it.

2.3 A Three-Level Framework of Research-Policy Alignment

This paper proposes that research-policy alignment be understood as a three-level construct rather than a binary property of shared labels:

Level 1: lexical or bibliometric correspondence. The dominant mode of existing SDG-mapping work: assigning texts to goals through keyword dictionaries, Boolean queries, or citation and mention counts (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024). This level is criticised: shared vocabulary does not guarantee shared meaning (Bornmann, Haunschild, and Marx, 2016), and keyword methods are brittle to vocabulary and database choices (Armitage, Lorenz, and Mikki, 2020).

Level 2: observed textual-semantic proximity in embedding space. Sentence embeddings locate texts in a shared semantic space where proximity reflects distributional patterns of co-occurrence (Harris, 1954; Reimers and Gurevych, 2019), and encoder-based SDG association has been benchmarked as a strong baseline (Gjorgjevikj et al., 2025). This level detects divergence that keyword methods miss. It also makes divergence structurally visible: observed proximity decomposes into separable components (coverage versus framing, and topic versus register within framing), and because the decomposition operates on the texts themselves, its findings can be reproduced and tested (Section 3.8). The present study is positioned at this level, and the limits of that measurement are developed in Section 2.4 below.

Level 3: substantive comparison of implied problems, actors, and interventions. Whether the two discourses address the same problems, involve the same actors, and propose the same interventions (Cash et al., 2003; Guston, 2001; Haas, 1992). Because this requires interpreting causal beliefs, legitimacy, and institutional mechanism (Caplan, 1979; Haas, 1992), it lies beyond unsupervised text analysis and is left to qualitative follow-up.

Advancing from Level 1 to Level 2 is the study's main methodological contribution. Level 3 is not addressed. Section 5.4 discusses a reference-relative precondition below these levels. Figure 1 summarises this decomposition.

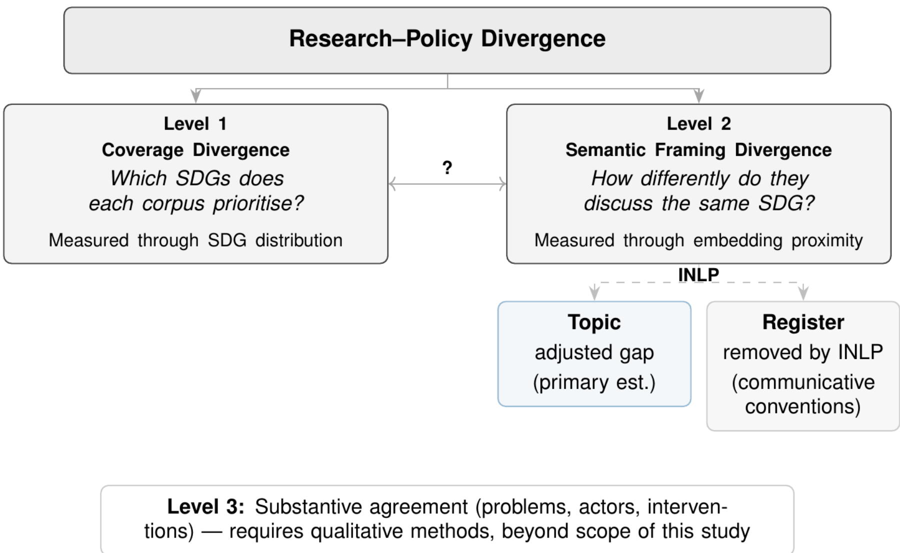

Figure 1. Conceptual framework: research–policy divergence separated into coverage and semantic framing dimensions, positioned within the three-level alignment construct.

2.4 Semantic Methods: Rationale and Cautions

The distributional hypothesis, that meaning can be described through patterns of linguistic co-occurrence (Harris, 1954), motivates the use of Sentence-BERT (Reimers and Gurevych, 2019) and related transformer encoders for large-scale semantic comparison. However, using them to compare research abstracts and policy documents is new and not yet directly benchmarked. The distance is descriptive rather than causal.

Gjorgjevikj et al. (2025) benchmarked sentence encoders for SDG association tasks, finding that general-purpose sentence-transformers models perform among the strongest baselines, and that domain-specific fine-tuning improves predictive performance while reducing sensitivity to description length. Rodriguez, Spirling, and Stewart (2023)'s conText framework provides a computational-social-science precedent for comparing meaning across contexts, adopting a narrow, structuralist view of meaning as distributional co-occurrence and making no causal claims about human understanding. This study follows that same descriptive premise: semantic proximity is a distributional measurement rather than evidence of substantive framing alignment. Because encoders also encode register and style (Grieve et al., 2025; Niu and Carpuat, 2017), embedding distance mixes topic, register, and convention into one number. This motivates the decomposition (Section 3.8).

McCarthy et al. (2026) supply a closer empirical precedent, showing with a multi-target SciBERT classifier that research-policy coverage gaps are measurable with transformer encoders outside the AI-SDG setting. This study extends that logic from a single conservation domain to all 17 SDGs, and from coverage to within-SDG semantic framing.

An older corpus-linguistic tradition adds a further caution. Biber's multidimensional (MD) analysis identifies six co-varying dimensions of linguistic variation across 23 spoken and written genres, derived from factor analysis of 67 linguistic features (Biber, 1988). Each dimension captures a continuum of co-occurring linguistic features rather than a binary speech/writing split (Biber, 1988; Biber and Conrad, 2009). Register is therefore a multi-layered property. INLP removes a linear slice isolating the largest direction of writing-style difference, and is not claimed to capture every aspect of register. For the present study this matters because research abstracts and policy documents may differ not only in substantive topic, but also in abstraction, modality, institutional stance, and rhetorical function. Embedding distance is therefore treated as observed textual-semantic proximity rather than a clean measure of topic alone.

2.5 Research Gap and Hypotheses

The literature reviewed above has not produced (i) a systematic comparison of how academic AI-for-sustainability research and policy discourse differ in SDG coverage and within-SDG semantic framing (existing work measures research coverage, citation patterns, or policy uptake without comparing the two corpora directly), or (ii) an examination of how coverage gaps and semantic gaps are related. This framework is diagnostic. It maps where observed divergence is, without classifying whether any given gap reflects productive or problematic disconnection.

The research question is operationalised by four sub-hypotheses, each correlating a distinct coverage predictor with the within-SDG semantic gap across the 17 SDGs:

- **H1a** (motivating hypothesis): the coverage gap is positively associated with the semantic gap;
- **H1b**: signed dominance (research share $-$ policy share) $\leftrightarrow$ semantic gap;
- **H1c**: research coverage $\leftrightarrow$ semantic gap;
- **H1d**: policy coverage $\leftrightarrow$ semantic gap.

H1b–H1d are signed tests of directional components: with dominance defined as research share minus policy share, the sign of each association locates which community's framing norms drive the linkage. Operationalisation (Pearson r and Spearman ρ between each predictor and the semantic-gap scores, $n = 17$ scope conditions) is described in Section 3.9.

These contributions and hypotheses are motivated by the theoretical frameworks reviewed in Section 2.2. The two-communities theory holds that researchers and policymakers occupy separate worlds with different values, reward systems, and languages (Caplan, 1979). If these communities also develop different topical priorities, as the empirical record of research-policy divergence suggests (Strauss et al., 2025; McCarthy et al., 2026), then coverage divergence and semantic divergence should co-occur: SDGs where one community dominates attention should also show divergent framing, because the community shaping the discussion brings its own communicative norms. The epistemic-communities framework further distinguishes the direction of dominance: where research prioritises a given SDG, the framing should reflect academic norms of knowledge production. Where policy prioritises it, the framing should reflect institutional norms of coordination (Haas, 1992). A raw comparison alone cannot distinguish co-occurring divergence from offsetting channels, which is what the decomposition of Section 3.8 is designed to adjudicate.

3 METHODOLOGY

This study uses publicly available text data and requires no ethical approval.

3.1 Token-Aware Segmentation

All source documents are segmented at sentence boundaries using the embedding model's (Section 3.2) own tokenizer to enforce a hard token budget. The procedure applies identically to research abstracts, policy documents, and reference corpora: NLTK sentence-boundary detection is followed by greedy accumulation of sentences up to a token budget of `max_seq_length` $-$ 10 (for the canonical MPNet encoder: 384 $-$ 10 = 374 tokens). Segments falling below 20 words are discarded.

For the research–policy comparison this design has two consequences. First, research abstracts (mean $\sim$200 words) mostly fit within the 374-token budget, but 17.2% exceed it and split into multiple segments. The research-side text unit is therefore the abstract, whose segments are aggregated to a single document vector before assignment. Second, policy documents of variable length are split into multiple segments of comparable size to research abstracts, making the two corpora directly comparable in embedding space.

3.2 Embedding Model and Normalisation

All texts were encoded with `all-mpnet-base-v2` (Sentence-Transformers, 2021b; Reimers and Gurevych, 2019) from the sentence-transformers library, producing 768-dimensional dense vectors. Embeddings were L2-normalised to unit vectors so that cosine similarity equals their dot product.

No fine-tuning is used. The embeddings are frozen. A linear classifier on frozen embeddings (linear probing) is the standard way to test what a representation already captures (Alain and Bengio, 2018; Conneau et al., 2018). Both the SDG classifier (Section 3.5) and the INLP step (Section 3.8) are linear operations on this fixed space. Freezing also keeps the two corpora in one shared space. Retraining on the SDG data (which skews toward policy words) would distort that space.

Encoder choice is a primary robustness axis: gap rankings are re-computed under two alternative encoders, all-MiniLM-L6-v2 (Sentence-Transformers, 2021a) (384-dimensional) and SciBERT (Beltagy, Lo, and Cohan, 2019; Allen Institute for AI, 2019) (768-dimensional). Encoder-robustness checks use the same reference (Section 3.5) and policy corpora (Section 3.4), but only 100,000 research abstracts drawn once with a fixed seed from the full research-abstract corpus (Section 3.3) to save computation, following evidence that the gap ranking stabilises near 50,000 abstracts under repeated resampling, while gap magnitudes stabilise by roughly 200,000 (Appendix E). The subset exceeds the corpora used in prior SDG-classification studies (Section 2.1).

3.3 Research Corpus

The research corpus consists of 2,536,771 research abstracts retrieved from the OpenAlex API (Priem, Piwowar, and Orr, 2022), covering publications between 2018 and 2025.

Retrieval combined OpenAlex's native SDG work filter with four high-precision AI search terms (`artificial intelligence`, `machine learning`, `deep learning`, `neural network`), giving 68 queries (17 SDGs $\times$ 4 terms). The term set draws on established practice for delimiting AI corpora (Hellwig, Huggett, and Siebert, 2019; OECD, 2024), capturing AI-related work without trying to catch every AI term. The search favours precision over recall: each term must mark AI work so that non-AI papers stay out of the embedding space, accepting misses. Schoeberl, Toney, and Dunham (2023) reach the same precision-over-recall conclusion for AI-research identification. The 'learning' terms plausibly inflate the SDG 4 share. This is examined in Appendix B.1.

A direct test re-ran retrieval using field-of-study membership for the same two AI fields/concepts of "AI" and "ML" (OECD, 2024) and re-scored the resulting 100,000-paper corpus through its own segmentation (Section 3.1) and classifier (Section 3.5), but reuses the register-adjusted space from the main research corpus (Section 3.8), assuming research-policy register directions are unchanged in the same semantic space. Results are reported in Section 4.4, with detailed diagnostics in Appendix A.

The robustness check applies no SDG filter at retrieval. SDG labels are assigned post hoc by the classifier, so the corpus is not independently verified as SDG-relevant (Section 5.5).

After deduplication and removal of records lacking usable abstracts and the 20-word minimum, the research corpus survives as 2,536,771 abstracts, tokenised into segments for embedding (record and segment counts in Table 1). The corpus is predominantly English (99.5%) but no hard language filter is applied. The main results reproduce under subsampling from 50,000 papers (Appendix E). No balancing by SDG or citation count is applied.

3.4 Policy Corpus

The policy corpus contains text segments from three collections, representing distinct source documents (segment and document counts per collection in Table 1). It should be read as mixed international institutional SDG policy discourse rather than every AI-sustainability policy that exists:

- **Curated AI and SDG policy documents**: UN, OECD, EU, UK, and AU AI governance frameworks, IPCC assessment reports, SDSN sustainable development reports, and national AI strategies.
- **SDGi Voluntary National Review and Voluntary Local Review corpus** (SDGi, 2024): National and local self-assessments submitted to the UN High-Level Political Forum.
- **UN General Debate Corpus** (Jankin, Baturo, and Dasandi, 2024): SDG-relevant passages from General Debate speeches, sessions 70–80 (2015–2024). Paragraph-level keyword filtering precedes segmentation, so only SDG-relevant passages enter the pipeline and its segment count is accordingly far smaller than the full speech corpus.

The SDGi VNR/VLR collection is also one of the five supervised reference corpora (Section 3.5), so its policy segments carry gold SDG labels. An asymmetry follows: the research corpus is an intersection of AI and SDG content while the policy corpus is a broader union covering institutional SDG discourse and AI governance. The curated sub-corpus is the more relevant reference for AI-governance-specific interpretation. Appendix H.7 reports a source-family sensitivity check.

3.5 Supervised Reference Classifier

The canonical instrument is a supervised classifier trained on pooled single-label reference data from five independent annotated corpora. Corpus sizes at each pipeline stage (raw, preprocessed + de-duplicated, segmented, and single-label) are reported in Table 1:

- **OSDG Community Dataset** (OSDG) (Pukelis et al., 2022) (no SDG 17): crowd-sourced excerpts validated by multiple volunteers.
- **SDG Classification Benchmark** (SDGCB) (SDG Classification Expert Group, 2025) (expert-verified): expert-annotated snippets with consensus resolution.
- **IISD SDG Knowledge Hub** (SDGKH) (Wulff, Meier, and Mata, 2023): multi-label journalism and policy commentary on 2030 Agenda implementation.

- **SDGi Corpus** (SDGi) (SDGi, 2024): government Voluntary National and Local Reviews (VNRs/LRs).
- **Aurora dataset** (Aurora) (Vanderfeesten, Spielberg, and Gunes, 2020) (expert-validated): academic abstracts validated by domain researchers.

Table 1. Corpus provenance: record counts at each pipeline stage.

Corpus	Raw	Prep	Seg	Lab
OSDG	43,025	30,532	30,532	30,532
SDGCB	1,251	281	281	281
SDGKH	9,172	9,171	26,959	6,122
SDGi	5,880	4,225	21,346	20,267
Aurora	5,619	4,022	4,975	4,971
Research (OpenAlex)	2,763,579	2,536,771	3,105,144	—
Policy (3 Collections)	6,368	6,367	40,597	—

Notes: Raw is source size before processing. Prep is cleaned, de-duplicated records surviving the 20-word minimum, i.e. documents yielding at least one segment. Seg is the post-segmentation segment count. Lab is the single-label count feeding the classifier (— for the unlabelled research and policy corpora). OSDG and SDGCB bypass segmentation, so Seg = Lab.

OSDG alone contributes about 49% of labelled texts and draws on UN-adjacent documents, so the reference data remain weighted toward policy vocabulary. OSDG and SDGCB bypass segmentation due to having short texts. SDGKH, SDGi, and Aurora are segmented before pooling.

The pooled reference data were split into training and held-out test sets by stratified document-grouped splitting (pool size and split are reported in Table 1), keeping all texts from the same source document in one fold. The champion LR ($C = 3.0$, L2 penalty, `class_weight=None`, solver `lbfgs`) was selected by 5-fold GroupKFold cross-validation (Hastie, Tibshirani, and Friedman, 2009) on the training pool. Appendix F gives the MLP robustness comparison.

The validation task does not evaluate a rejection option for non-SDG texts. This is acceptable for the primary corpus, which is restricted to an OpenAlex AI-and-SDG universe so the classifier answers a relative question among the 17 SDGs. The concept-retrieved robustness corpus lifts that restriction (Section 5.5).

Each text is assigned to its highest-probability SDG (argmax), traceable to 768 signed coefficients per class with no hidden layers, no dropout, and no random-seed sensitivity. Macro-F1, the unweighted mean of per-SDG scores, is the primary validation metric (See results in Section 4.1).

Because the classifier is trained on pooled reference data that are themselves policy-adjacent, the scoring instrument behaves differently on the two corpora. Policy segments receive systematically higher LR predicted probabilities than research abstracts (mean 0.749 vs 0.627). The gap has two causes the data cannot separate: a genre artefact (the reference data skew toward policy vocabulary) or a substantive difference (policy texts address SDG implementation more directly). Hard argmax assignment depends only on each text's within-corpus probability ranking across the 17 SDGs, so the asymmetry threatens the findings only if it shifts SDGs unevenly within a corpus.

Two supervised model families were compared: logistic regression, which exposes interpretable per-dimension coefficients, and a multi-layer perceptron (MLP). Random forests and gradient boosting were excluded because the embedding space is already learned, so bagged trees add ensemble complexity without interpretable gain.

The LR grid searched $C \in \{0.1, 1, 2, 3, 5, 10, 30, 100\}$ with an L2 penalty and `class_weight` $\in$ {`None`, `balanced`}. The MLP grid swept $n_{\text{layers}} \in \{2, 3, 4, 6\} \times$ hidden_size $\in \{256, 384\} \times$ lr $\in \{3\text{e}{-}4, 1\text{e}{-}3\}$ with `weight_decay`=0, `dropout`=0.3, AdamW, and early stopping.

For robustness, a zero-shot nearest-centroid method (hereafter ZS) assigns each unlabelled text to the nearest reference-SDG centroid in the embedding space.

Grid search and test results, and the resulting choice of LR, are reported in Section 4.1.

3.6 Coverage Gap Analysis

For each corpus item, the predicted SDG is the LR argmax (hard assignment, Section 3.5). The coverage profile is the proportion of analysis units assigned to each SDG (an abstract for research, a source document for policy).

The policy corpus comprises 6,367 source documents of different lengths (40,597 segments in total, Section 3.4). A segment-weighted profile would let a few long reports dominate, so the profile is document-weighted: every source document counts equally, contributing one length-normalised unit vector (Manning, Raghavan, and Schütze, 2008). In practice, segment vectors are averaged per document, hard-assigned to the top SDG, and the coverage profile counts documents per SDG. The research side follows the same logic at paper level, so both corpora weight units identically.

The coverage gap for SDG j is defined as:

$$\text{CoverageGap}_j = \left|\text{ResearchCoverage}_j - \text{PolicyCoverage}_j\right|$$

The signed version of the same quantity is:

$$\text{SignedCoverageGap}_j = \text{ResearchCoverage}_j - \text{PolicyCoverage}_j$$

A positive value means research dominates SDG j. A negative value means policy dominates.

3.7 Semantic Gap Analysis

The semantic gap measures within-SDG divergence: given that both corpora have texts assigned to SDG j, how differently do they discuss it?

For each SDG j the analysis collects all assigned research papers and all assigned policy segments (capped at 50 per source document to bound single-document influence). Unlike the document-weighted coverage profile (Section 3.6), aggregation differs across the two corpora here by design. Research abstracts are single-topic by genre convention, so each of the 2,536,771 abstracts contributes one unit vector, the L2-renormalised mean of its segment embeddings, and a 33-segment abstract carries equal weight to a single-segment one: the research centroid $\mathbf{c}_j^{\text{res}}$ is the L2-normalised mean of these paper-level vectors assigned to SDG j. Policy sources are long, deliberately excerpted institutional texts (UNGDC speeches, SDGi VNRs) whose passages are topically distinct and may span several SDGs, so equal document weighting would collapse that within-document variation. The policy centroid $\mathbf{c}_j^{\text{pol}}$ is therefore the L2-normalised mean of the (capped) segment embeddings assigned to SDG j, sampling each passage's topical content.

The semantic gap for each SDG is then defined as:

$$\text{SemanticGap}_j = 1 - (\mathbf{c}_j^{\text{res}} \cdot \mathbf{c}_j^{\text{pol}})$$

A gap of 0 indicates perfectly overlapping semantic directions. A gap approaching 1 indicates orthogonal framing. Cosine similarity between centroids is the standard method for document-level semantic comparison in embedding space (Brück and Pouly, 2019; Babićc et al., 2020). Centroid cosine distance collapses each embedding cloud to its mean direction.

Appendix D compares the adjusted-gap ranking against full-distribution alternatives, with moderate agreement for most metrics and weak or negative agreement for the rest. An alternative reading treats the weak agreement as evidence against the mean-direction centroid. The study cannot separate the two explanations and reports both distribution-aware and centroid results.

The per-document segment cap of $K = 50$ (seeded, reproducible) mitigates the influence of long reports on the policy centroid. Sensitivity at $K = 20$ and no cap is reported in Table 17, Appendix H.

3.8 Register Adjustment via Iterative Nullspace Projection (INLP)

The semantic gap is not a clean measure of substantive disagreement. Research and policy texts differ in both topic (what they discuss) and register (how they are written – audience, purpose, style). Embedding distance mixes the two. INLP (Ravfogel et al., 2020) is used in this paper to split the gap into an adjusted semantic gap (topic) and a register component (See results in Section 4.3).

Objective: INLP learns directions that, when subtracted from the embeddings, erase the research–policy difference. After K iterations, held-out corpus-classification accuracy is at chance on the stratified training distribution. A residual corpus signal remains detectable out of sample (Appendix G.2). Each iteration removes the direction carrying the most remaining corpus signal.

SDG-stratified procedure: Vanilla INLP removes a binary attribute (e.g. gender). The present adaptation must remove register, a corpus-level property confounded with SDG topic in the raw embeddings. The modification is SDG stratification: at each iteration, equal numbers of research and policy embeddings are drawn from each of the 17 SDGs, so the classifier cannot use topic as a stand-in for register. The procedure proceeds as follows at each iteration $k = 1, \ldots, K$:

1. **Stratified sample:** Independently redraw n_{target} texts per SDG per corpus without replacement, under the global-minimum rule $n_{\text{target}} = \min_j \min(|X_j^{\text{res}}|, |X_j^{\text{pol}}|)$ ($n_{\text{target}} = 1123$ for MPNet on the frozen snapshot). Form the 34-class stratification key $z_{\text{strat}} = 2 \times$ sdg_labels $+\, y$, ensuring balanced corpus $\times$ topic combinations.
2. **Project existing directions:** Subtract from each sample embedding the components along all previously learned directions: $\tilde{\mathbf{x}} = \mathbf{x} - \sum_{j<k} (\mathbf{g}_j \cdot \mathbf{x})\, \mathbf{g}_j$.
3. **Train classifier:** Fit L2-regularised logistic regression to predict corpus identity y from the projected embeddings $\tilde{\mathbf{x}}$. The hyperparameters are fixed defaults because the objective is to recover a decodable direction, not to maximise classification accuracy. No per-iteration direction check is performed. The aggregate stopping criterion (held-out accuracy ≤ 0.5 across all K iterations) guards against retaining

a degenerate or noisy direction. Split 15% held-out, stratified by z_{strat}.

4. **Extract direction:** Normalise the weight vector: $\mathbf{g}_k = \mathbf{w}_k / \|\mathbf{w}_k\|$. Orthogonalise against all prior directions: $\mathbf{g}_k \leftarrow \mathbf{g}_k - \sum_{j<k} (\mathbf{g}_k \cdot \mathbf{g}_j)\, \mathbf{g}_j$. If $\|\mathbf{g}_k\| < 10^{-8}$ (direction collapse), stop.
5. **Stopping criterion:** Stop if held-out accuracy ≤ 0.5 (the majority-class baseline for balanced binary classification). Otherwise append $\mathbf{g}_k$ and continue.

All three encoder tracks converge within 40–79 iterations (Appendix G.1). The final adjusted embedding is $\mathbf{x}' = \mathbf{x} - \sum_{k=1}^{K} (\mathbf{g}_k \cdot \mathbf{x})\, \mathbf{g}_k$, and the adjusted gap uses centroids computed on these projected embeddings.

Identification argument: The stratification design is the structural precondition: no linear direction can simultaneously encode SDG topic and corpus identity. What remains linearly decodable is the corpus-level register difference: modality, audience conventions, institutional stance, and structural patterns. The removed subspace is interpreted as register in the sense of Biber and Conrad (2009): a variety associated with a particular situation of use, including communicative purpose, audience, and production circumstances. Stratification controls for SDG topic, but not for other differences such as publication period or author background. The removed component is therefore best read as register-like. Register is partly defined by topic and domain (Grieve et al., 2025), so cross-cutting development/economics prevalence is not excluded by construction; P1/P2 show the component is not SDG topic. This interpretation is evaluated against independent linguistic markers in Appendix G.2.

With the projector learned, the semantic-gap analysis of Section 3.7 is recomputed on the adjusted embeddings, giving the within-SDG adjusted gap.

3.9 Coverage–Semantic Interaction

The test works as follows. Pearson r and Spearman rank correlation ρ are computed between each coverage predictor (coverage gap, signed dominance, research coverage, policy coverage) and the within-SDG semantic-gap scores across the 17 SDGs. SDG 4 is treated as artefact-affected (Section 3.3) and examined separately (Appendix B.1).

A possible criticism is that $n = 17$ limits statistical power. However, the 17 SDGs are the whole framework, not a sample. Finer-grained units (the framework defines 169 targets) would raise the observation count, but no labelled target-level reference data exist. That is left for future work when data is available.

The only randomness comes from measurement (classifier, corpus build, embeddings). All p-values are two-sided Monte Carlo permutation tests (100,000 resamples, fixed seed). They test the null of no association given the measured values, without relying on asymptotic approximations (Berk, Western, and Weiss, 1995; Abadie et al., 2020).

With only 17 units, a single goal can sway the result, so the associations are checked with a leave-one-out analysis (Section 4.4) and across configurations (Section 4.5).

Raw, adjusted, and register-component correlations come from one fixed decomposition: the register component equals raw minus adjusted for every SDG by construction, so the three correlations are linked rather than independent tests. No multiple-comparison correction is applied to them, and this reasoning covers the raw/adjusted/register triad only. It does not extend to the four H1a–H1d hypotheses or the $\sim$24 robustness configurations reported elsewhere, which are interpreted collectively for consistency of sign and magnitude. All three gaps are reported for every configuration. The separate evidence for the register reading comes from Appendix G.2 itself.

A supplementary pooled-regression analysis combines all 24 configurations into a single panel (Appendix I).

Figure 2 summarises the end-to-end methodology.

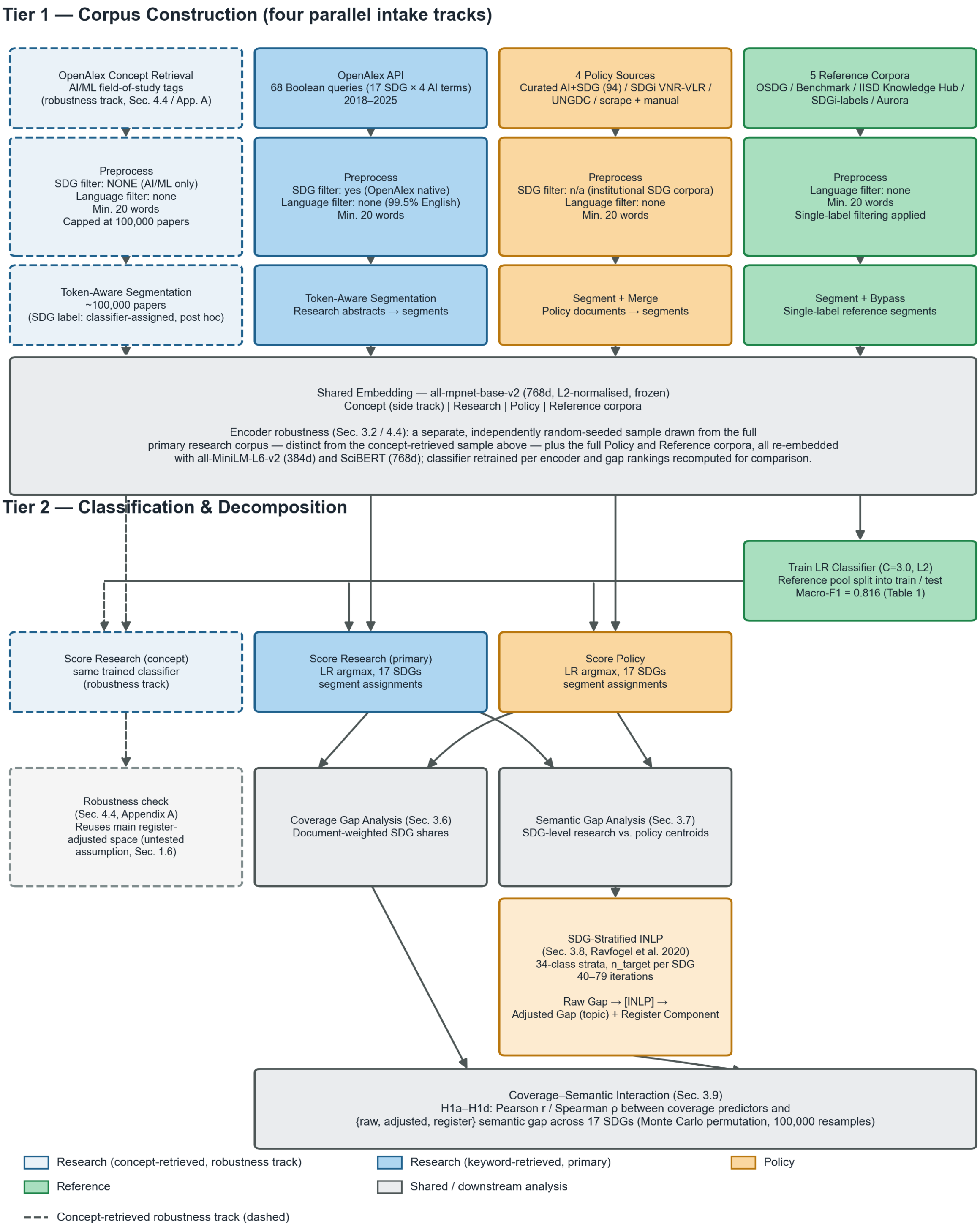

Figure 2. End-to-end methodology pipeline.

4 RESULTS

4.1 Supervised Reference Classifier Grid Search and Performance

Grid search over architecture depth, hidden size, and learning rate showed negligible performance variance across the MLP configurations (CV macro-F1 = 0.815–0.819), and LR is flat within $\sim$0.001 for C $\in \{1, 3, 5\}$, suggesting that LR/MLP choice is not a major binding constraint (Table 2).

Table 2. Model selection ranking: 5-fold GroupKFold cross-validated macro-F1.

Rank	Config	CV macro-F1
1	MLP 3L/256h, lr=0.0003 (best)	0.8192 $\pm$ 0.0032
2	MLP 2L/256h, lr=0.0003	0.8184 $\pm$ 0.0030
3	MLP 6L/384h, lr=0.0003	0.8183 $\pm$ 0.0023
4–16	MLP (remaining 13 configurations)	0.8151–0.8183
17	LR C=3.0, cw=None (chosen)	0.8101 $\pm$ 0.0020
18	LR C=2.0, cw=None	0.8098 $\pm$ 0.0018
19	LR C=5.0, cw=None	0.8097 $\pm$ 0.0017
20–32	LR (remaining 13 configurations)	0.7904–0.8087

Notes: Top three configurations per family shown, others collapsed to their CV macro-F1 range. LR (C=3.0, L2, class_weight=None) is the champion (rank 17).

LR (C=3.0, L2, class_weight=None) was chosen over the highest-scoring MLP despite a 0.0091 macro-F1 gap: LR requires one free parameter (C) and no hidden layers, dropout, learning-rate tuning, or early stopping, and it provides 768 signed coefficients per SDG class, each directly attributable to a specific semantic dimension, whereas hidden layers diffuse this attribution across non-linear transformations. For building an interpretable supervised reference to compare corpora, that transparency justifies accepting the lower score.

The chosen classifier was evaluated on the held-out test set (n=9,338, document-grouped to prevent source leakage). It reaches macro-F1 = 0.816 and micro-F1 = 0.8306, far above the 1/17 random baseline (0.059). Per-class F1 values from the test set are reported in Table 3.

Per-class F1 ranges from 0.610 (SDG 10) to 0.903 (SDG 3). Most goals score between 0.76 and 0.89. The weakest classes (SDGs 10 and 8) match the semantic breadth reported by Kashnitsky et al. (2024) (Appendix B.2, Figure 8).

Despite OSDG (49% of training data) having zero SDG 17 examples (Section 3.5), performance does not degrade (Table 3, Appendix B.2), which suggests the other four sources (SDGi, Aurora, SDGCB, SDGKH) carry enough SDG 17 signal to compensate.

Table 3. LR, MLP, and zero-shot (nearest-centroid) test-set F1 scores (n=9,338).

SDG	n (test)	LR test F1	MLP test F1	ZS test F1
1 (No Poverty)	473	0.737	0.758	0.671
2 (Zero Hunger)	416	0.830	0.829	0.792
3 (Good Health)	848	0.903	0.909	0.871
4 (Education)	734	0.889	0.899	0.878
5 (Gender Equality)	816	0.881	0.882	0.833
6 (Clean Water)	552	0.879	0.877	0.849
7 (Clean Energy)	602	0.861	0.862	0.845
8 (Decent Work)	398	0.716	0.718	0.631
9 (Innovation)	526	0.770	0.787	0.694
10 (Reduced Ineq.)	423	0.610	0.644	0.485
11 (Sust. Cities)	451	0.762	0.786	0.690
12 (Consumption)	330	0.783	0.775	0.720
13 (Climate)	564	0.831	0.845	0.778
14 (Life Below Water)	327	0.892	0.895	0.864
15 (Life on Land)	545	0.865	0.864	0.836
16 (Peace & Justice)	798	0.879	0.896	0.806
17 (Partnerships)	535	0.788	0.814	0.671
Macro-F1 (SDGs 1–17)	–	0.816	0.826	0.760
Micro-F1 (SDGs 1–17)	–	0.831	0.840	0.774

Notes: All three methods are evaluated on the same held-out pooled test set (all reference sources, document-grouped). The n (test) column is the per-SDG gold-label count in the test split, identical for every classifier. LR (C=3.0, L2) is canonical. MLP (3-layer/256-hidden) and zero-shot (raw nearest-centroid) F1s are comparison values, with commentary deferred to Appendix F.

A linear decision boundary suffices here: nearest-centroid assignment underfits it, while the MLP adds capacity without meaningful accuracy gain.

4.2 Coverage and Coverage Gap

Figure 3 presents the document-weighted policy coverage profile alongside the research coverage profile.

Policy corpus: Three SDGs dominate the document-weighted profile: SDG 16 (Peace, Justice and Strong Institutions, 17.3%), SDG 17 (Partnerships for the Goals, 11.7%), and SDG 13 (Climate Action, 11.3%), accounting for 40.3% of policy documents. This ordering is consistent with the literature review's account of SDG 17 as the cross-cutting means-of-implementation goal (Section 2) (Le Blanc, 2015; Griggs et al., 2017). The full-corpus profile is source-family sensitive (Appendix H.7): curated AI/SDG documents skew innovation-heavy, UNGDC speeches concentrate on climate, partnerships, and governance, and SDGi reviews are comparatively even.

Research corpus: Research is led by SDG 3 (25.7%), SDG 9 (20.1%), and SDG 4 (15.6%), then SDG 16 and SDG 11 (11.2%/5.1%). SDG 7 ranks sixth (4.1%), below the

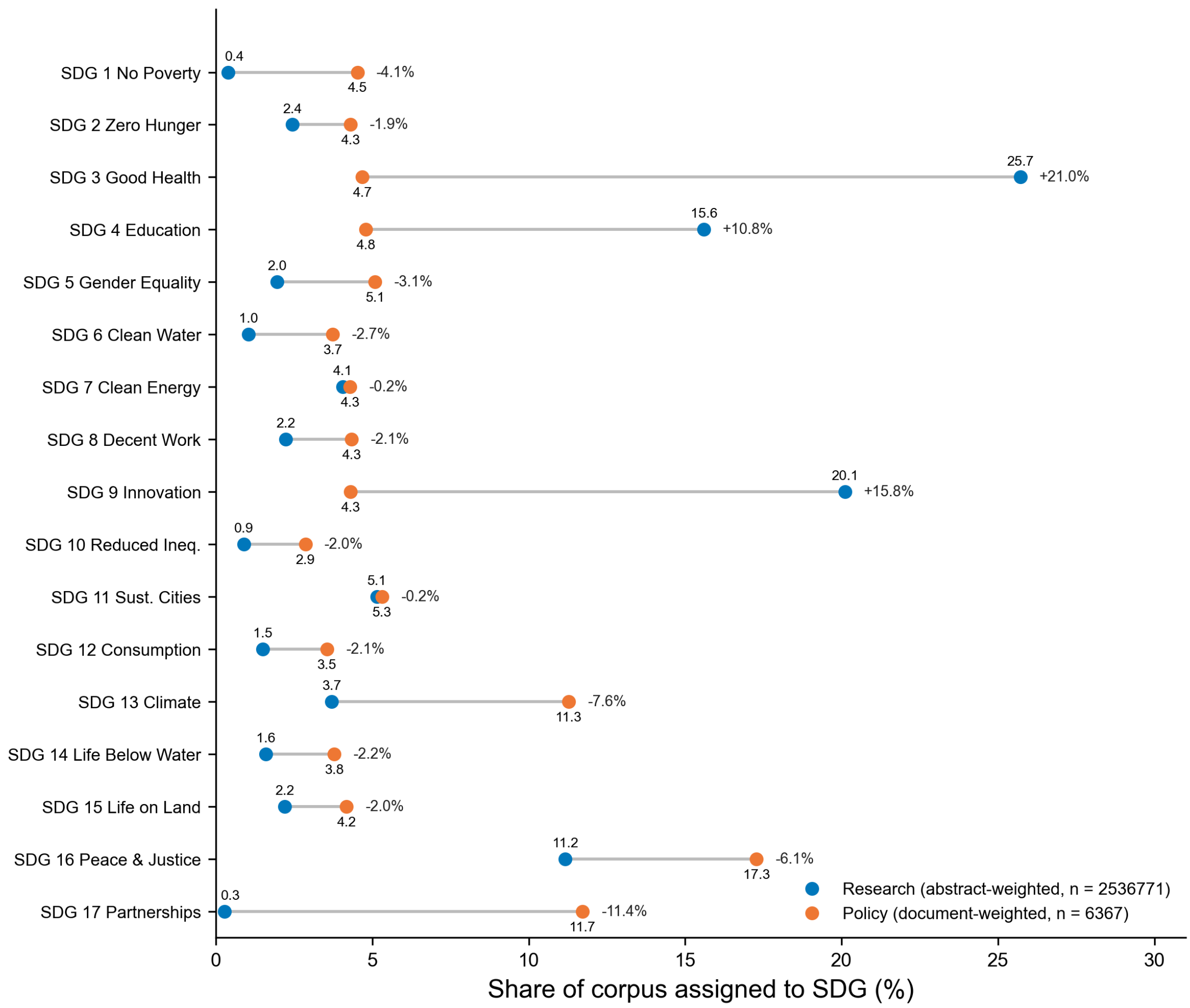

Figure 3. Coverage profiles by SDG.

joint lead position prior inventories report for it (Section 2.1). The SDG 3 lead and the SDG 4/11/16 pattern match prior inventories (Gohr et al., 2025). SDG 4 is likely inflated by retrieval and SDG 9 by the AI focus (Section 3.3). The near-absence of SDG 17 (0.3%) and SDG 1 (0.4%) matches Gohr et al. (2025).

Largest coverage mismatches: SDG 3 (21.0), SDG 9 (15.8), SDG 17 (11.4), SDG 4 (10.8), SDG 13 (7.6). Research dominates SDG 3 (25.7% vs. 4.7%), SDG 9, and SDG 4. Policy dominates SDG 17 (11.7% vs. 0.3%) and SDG 13 (11.3% vs. 3.7%).

4.3 Register Adjustment and Semantic Gap

Register adjustment converges after 62 iterations for the MPNet embedding space. Iterations until convergence for all three encoder spaces are reported in Appendix G.1. Figure 4 shows the combined embedding space and the geometric effect of INLP register adjustment: the raw space separates research and policy clouds, while the adjusted space shows no major separation between them. Although the PCA projection is coarse, the adjusted space removes the research–policy separation while leaving SDG separability intact (Appendix G.2).

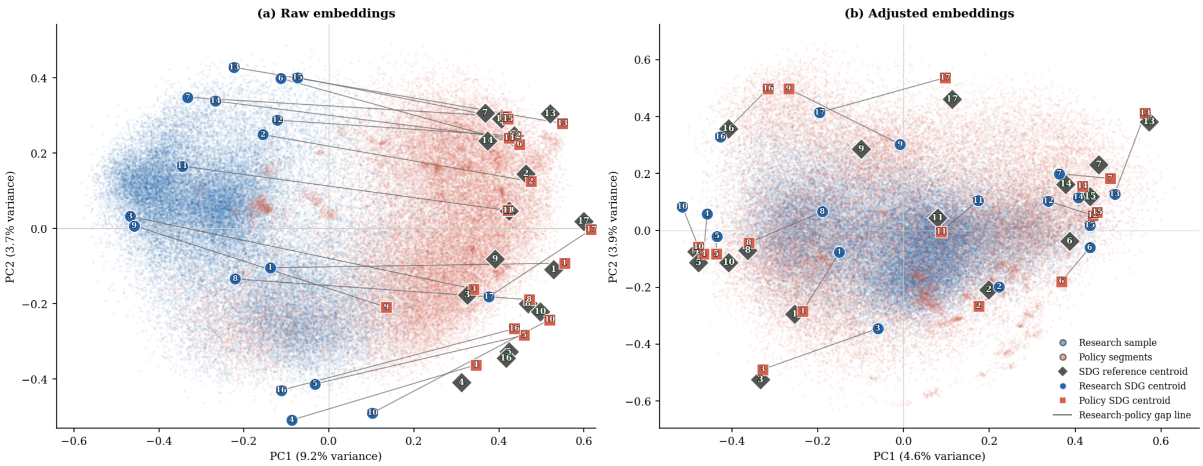

Figure 4. PCA before/after INLP register removal (MPNet).

Figure 5 shows the within-SDG gap sorted by the adjusted topic gap. The adjusted gap ranges 0.326–0.084 (SDG 17 widest, SDG 15 narrowest). The raw gap spans 0.494–0.177, widest at SDG 3. Both rankings are stable to a 20-or-none segment cap (Table 17), and size imbalance (398:1) does not drive them: on 10k/50k research draws the ranking reproduces at $\rho = 0.936/0.986$ (Appendix E).

Largest semantic gaps (adjusted): SDGs 17, 3, 16, 9, and 13 show the widest adjusted gaps. Under the raw gap, the ordering shifts: SDG 3 (0.494) leads. Register inflates the raw gap for 16 of the 17 SDGs. SDG 17 is the exception, where register increases apparent similarity.

Smallest semantic gaps (adjusted): SDG 15 (Life on Land, 0.084), SDG 2 (Zero Hunger, 0.089), SDG 6 (Clean Water, 0.130), SDG 14 (Life Below Water, 0.132), and SDG 5 (Gender Equality, 0.151) show the narrowest adjusted gaps. A narrow gap does not mean balanced attention: SDG 9 is still research-dominant, and SDG 4 carries the learning-vocabulary concern (Appendix B.1). Appendix C lists the most distinctive research- and policy-side terms for each SDG.

Figures 3 and 5 report the per-SDG coverage-predictor and semantic-gap values underlying the interaction tests.

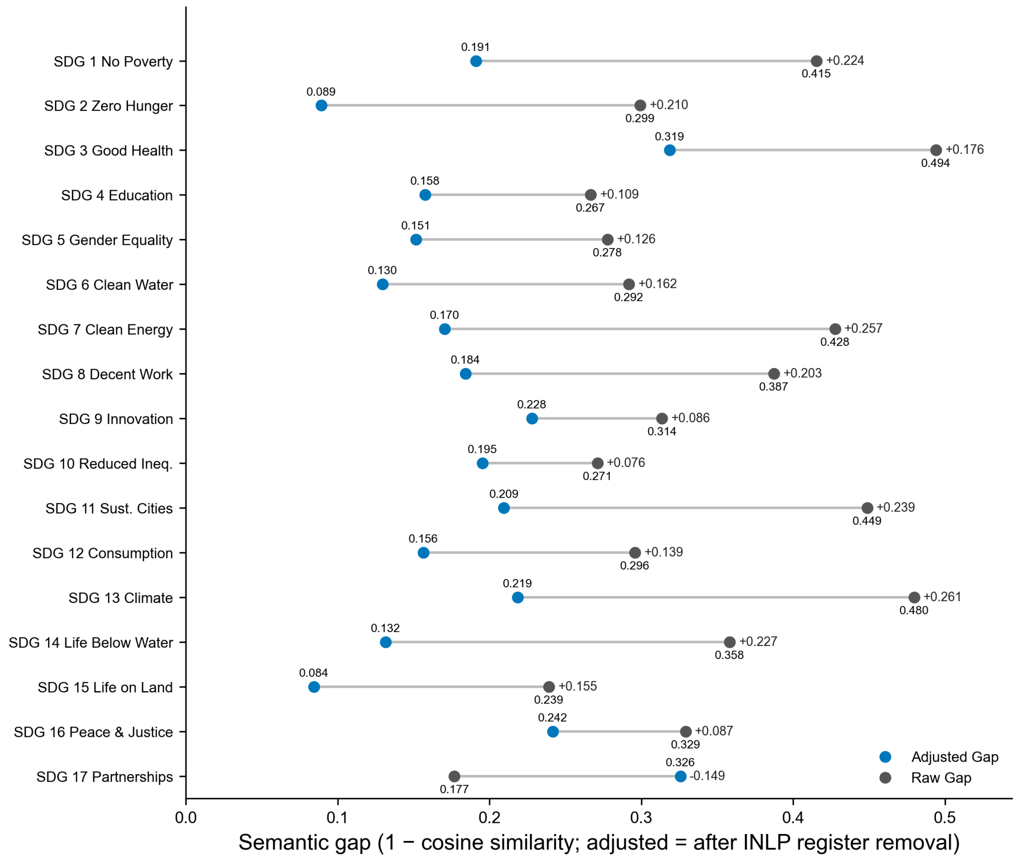

Figure 5. Within-SDG semantic gap by SDG, sorted by the adjusted (topic) gap after register removal.

4.4 Coverage–Semantic Interaction: Results

Table 4 reports H1a–H1d rank correlations between coverage predictors and the within-SDG semantic gap across all encoder–classifier configs. Columns show Spearman ρ against the raw gap, adjusted gap, and register component.

MLP configs reuse their encoder's INLP matrix. Concept rows use the concept-retrieved research corpus (Appendix A).

H1a is null on the raw gap ($\rho = -0.012$, $p = 0.959$; MPNet LR). The decomposition explains it: the register component is negatively associated ($\rho = -0.390$, $p = 0.120$) while the adjusted gap is positively associated ($\rho = 0.544$, $p = 0.025$), positive in 8/9 configs. The linkage is asymmetric: H1d (policy coverage) is positive in 9/9 configs (MPNet LR $\rho = +0.589$, SciBERT LR $\rho = +0.605$, Concept LR $\rho = +0.510$), while H1b/H1c show no consistent adjusted-direction signal.

The adjusted association is marginal: under a conservative familywise view of the three

Table 4. H1a–H1d coverage-predictor vs semantic-gap rank correlations (Spearman ρ) across encoder–classifier configs.

	Raw Gap	Adj. Gap	Reg. Comp.
H1a. Coverage gap $\leftrightarrow$ Semantic gap			
MPNet LR	-0.012	+0.544*	-0.390
MPNet MLP	-0.135	+0.336	-0.370
MPNet ZS	-0.299	+0.203	-0.449†
MiniLM LR	-0.243	+0.397	-0.463†
MiniLM MLP	-0.042	+0.505*	-0.272
SciBERT LR	-0.130	+0.221	-0.267
SciBERT MLP	-0.333	-0.034	-0.120
Concept LR	+0.108	+0.434†	-0.194
Concept MLP	-0.017	+0.181	-0.140
H1b. Research–policy dominance $\leftrightarrow$ Semantic gap			
MPNet LR	+0.159	-0.059	+0.029
MPNet MLP	+0.020	-0.118	-0.034
MPNet ZS	-0.005	-0.336	+0.360
MiniLM LR	+0.404	-0.020	+0.429†
MiniLM MLP	+0.319	-0.066	+0.348
SciBERT LR	+0.549*	-0.118	+0.547*
SciBERT MLP	+0.336	-0.429†	+0.529*
Concept LR	+0.051	-0.029	-0.260
Concept MLP	-0.174	+0.093	-0.199
H1c. Research coverage $\leftrightarrow$ Semantic gap			
MPNet LR	+0.449†	+0.267	+0.145
MPNet MLP	+0.262	+0.098	+0.088
MPNet ZS	-0.100	-0.328	+0.314
MiniLM LR	+0.444†	+0.262	+0.331
MiniLM MLP	+0.373	+0.275	+0.279
SciBERT LR	+0.605*	+0.466†	+0.020
SciBERT MLP	-0.005	-0.196	+0.225
Concept LR	+0.221	+0.017	-0.076
Concept MLP	-0.098	-0.010	-0.127
H1d. Policy coverage $\leftrightarrow$ Semantic gap			
MPNet LR	+0.212	+0.589*	-0.009
MPNet MLP	+0.175	+0.428†	-0.007
MPNet ZS	-0.094	+0.240	-0.321
MiniLM LR	-0.015	+0.449†	-0.159
MiniLM MLP	+0.083	+0.456†	-0.093
SciBERT LR	-0.100	+0.605*	-0.686**
SciBERT MLP	-0.172	+0.395	-0.395
Concept LR	+0.369	+0.510*	+0.178
Concept MLP	+0.256	+0.258	+0.082

Notes: Raw gap = naive semantic-gap estimate. Adj. gap = after INLP register removal (primary estimate, Appendix G.2). Register = raw $-$ adjusted. Each correlation is between two rank vectors (ranks 1–17). $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$, $^{\dagger}p < .10$.

linked statistics, $p = 0.025$ would not survive correction. It is read as pattern-consistent support for the offsetting decomposition, with the weight carried by cross-configuration and cross-encoder consistency (Section 4.5).

A metric-sensitivity check repeats this grid with Pearson r on the raw magnitudes instead of Spearman ρ (Appendix H.6). The adjusted-gap association is unchanged in sign under the metric switch for the signal-bearing hypotheses (H1a, H1d). The few sign changes sit in insignificant configs of the null H1b/H1c, confirming the rank-based reporting is not driving the result.

The SDG 4 lexical artefact (Appendix B.1) does not drive this result. A leave-one-out check with SDG 4 removed leaves the raw coverage-gap correlation null ($\rho = 0.088$, $p = 0.744$) and preserves the positive register-adjusted coverage-gap association ($\rho = 0.597$, $p = 0.016$), so the artefact is responsible for neither the raw null nor the adjusted association.

Restricting the policy side to the curated AI/SDG family (symmetric scope) preserves the raw-null (research share uncorrelated, coverage gap null). On register-removed gaps the curated family is null ($\rho = -0.061$, $p = 0.807$) while SDGi and the full corpus stay positive ($\rho = 0.498$, $p = 0.044$, Table 28, Appendix H.7).

Figure 6 plots the within-SDG semantic gap against each coverage predictor. Large coverage gaps span the full range of adjusted gaps (panel a), so no single SDG drives the near-zero raw association. The directional panels mirror the asymmetry above: no consistent trend for H1b or H1c, rising H1d.

4.5 Robustness of the Gap Rankings

Register removal is validated against independent linguistic markers (Appendix G.2).

Cross-sensitivity (policy source, segment cap, retrieval): Under MPNet–LR, the semantic-gap ranking is stable across segment cap and retrieval in the adjusted panel ($\rho \geq 0.65$, Table 17) and across policy source, segment cap, and retrieval in the raw panel ($\rho \geq 0.65$, Table 18). The same holds for coverage (Table 22).

Encoder sensitivity: Coverage-gap ranks are stable across encoders (MPNet–LR vs. SciBERT–LR $\rho = 0.934$; vs. MiniLM–LR $\rho = 0.895$; Table 21). Semantic-gap magnitude is encoder-dependent (SciBERT: 0.043 vs. MPNet: 0.340) because scientific pretraining fuses the genres (Beltagy, Lo, and Cohan, 2019; Allen Institute for AI, 2019), and the divergence conclusion holds (Table 19).

Classifier sensitivity: LR and MLP show varied agreement (MPNet LR vs. MLP $\rho = 0.850$, Table 19), with SDG 17 the most classifier-sensitive goal.

Distance-functional robustness: Most distribution-aware metrics correlate only moderately with the adjusted semantic gap (sliced Wasserstein $\rho = 0.525$, Chamfer $\rho = 0.485$), while Gaussian-2-Wasserstein and squared MMD are weaker ($\rho \approx 0.24$–0.26) and C2ST and Grassmann show negative agreement ($\rho = -0.164$ and -0.243, Appendix D). The adjusted gap is primarily a mean-direction finding. Shape carries a related but distinct signal.

Synthesis: Two conclusions survive: SDG 15 is robustly least-divergent across configs (adjusted rank 17). SDG 17 is most classifier-sensitive and most divergent under the adjusted gap, while SDG 3 leads under the raw gap for MPNet–LR. The raw-null pattern replicates across all configs and policy-source families. Pooled OLS confirms the adjusted association (Appendix I).

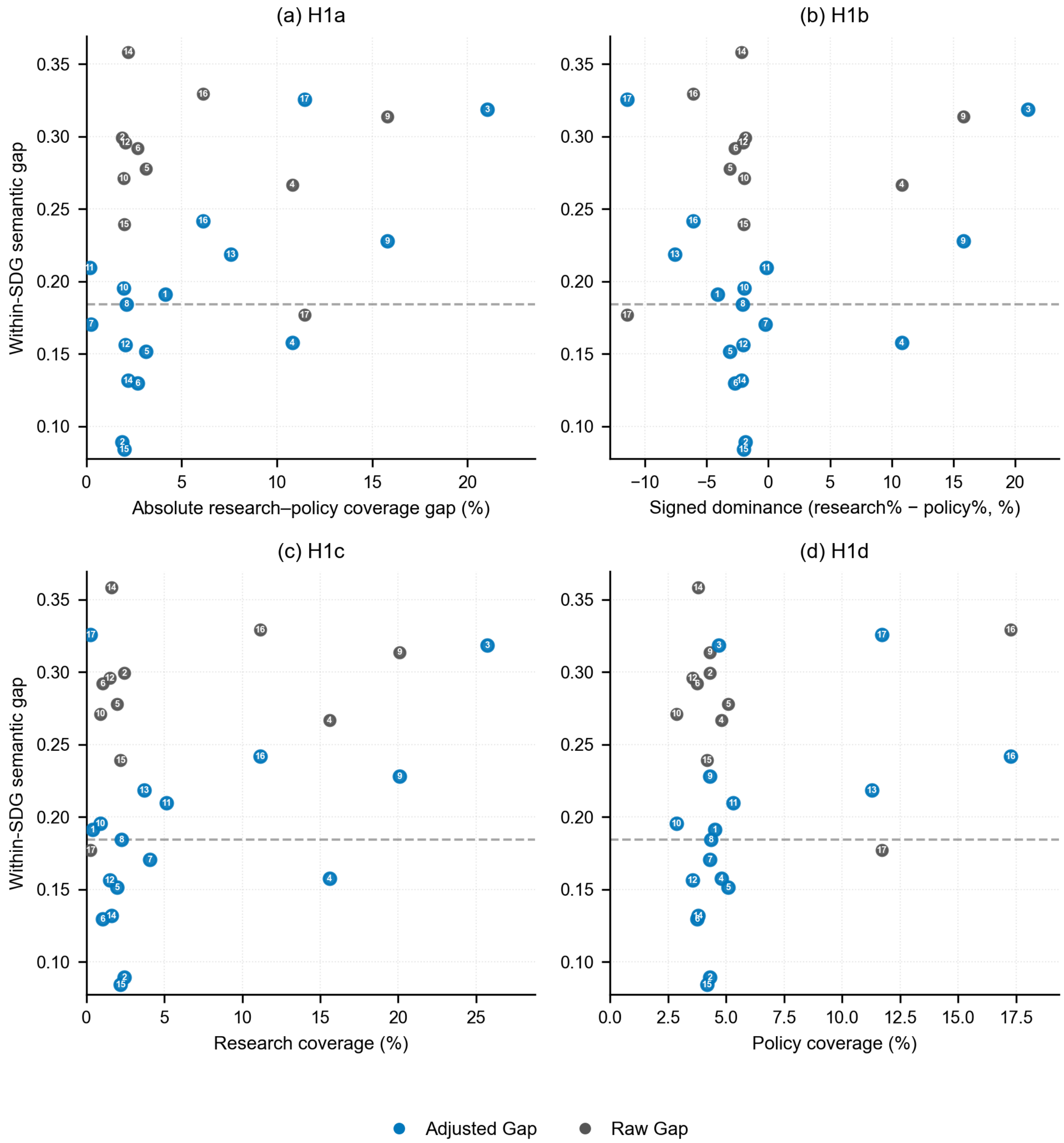

Figure 6. Coverage predictor vs within-SDG semantic gap, one point per SDG.
Notes: Semantic gaps from Figure 5. Coverage predictors from Figure 3. Cov. = |research − policy| (pp). Signed dom. = research share − policy share (pp).

5 DISCUSSION

The raw coverage-gap correlation is near zero ($\rho = -0.012$, $p = 0.959$). The adjusted gap is positively associated with coverage ($\rho = 0.544$, $p = 0.025$), whereas the register component has a negative association ($\rho = -0.390$, $p = 0.120$, Table 4).

5.1 Semantic cancellation effect

Independence of classification and measurement: Classification uses softmax probabilities. Semantic measurement uses cosine distance between L2-normalised centroids. The main confounder risk is INLP register-adjustment adjusting less for low-coverage SDGs than high-coverage ones, manufacturing rather than revealing the adjusted-gap correlation. Stratification rules this out (Section 3.8): at every iteration INLP draws an equal, globally fixed number of research and policy embeddings per SDG (n_target = 1123 for MPNet), regardless of actual coverage share. Coverage-as-label-share cannot therefore leak into the learned or removed direction.

Cross-configuration replication: This pattern replicates across most configs (Table 4): signs are stable for all encoder–classifier combinations under the primary keyword-retrieved corpus. Excluding the concept-retrieved track (Section 5.5), it does not appear to be a quirk of a specific encoder space, classifier, or methodological choice (segmentation, segment-capping for long documents).

SciBERT compression supports the register interpretation: SciBERT compresses the raw gap from 0.340 to 0.043 (88% reduction) while preserving coverage ($\rho = 0.934$; Table 21). Scientific pretraining (Beltagy, Lo, and Cohan, 2019; Allen Institute for AI, 2019) fuses the registers because both use scientific conventions, which the decomposition expects.

5.2 Two communities, two dimensions

If the cancellation is real, it calls for a theoretical explanation. The two-communities theory (Caplan, 1979) provides one. Caplan characterises social scientists and policy makers as living in “separate worlds with different and often conflicting values, different reward systems, and different languages” (Caplan, 1979, p. 459). The gap, he argues, “necessarily involves value and ideological dimensions as well as technical ones” (Caplan, 1979, p. 461). If the two communities are distinct knowledge cultures, then a near-zero raw correlation is expected: the two corpora develop different topical priorities (producing coverage divergence) and different communicative norms (producing register divergence), and the two effects operate in opposite directions on the semantic gap.

This account was formed after seeing the raw result, so it is judged by the commitments it enforces rather than by fit alone (Lakatos, 1970; Hitchcock and Sober, 2004). Those commitments held: topic selectivity survived removal (Appendix G.2), the adjusted association reproduced across supervised configurations (Table 4), and scientific-domain pretraining compressed the raw gap while preserving which goals diverge most (Section 4.5). It would fail if a native register adjustment within the symmetric AI-and-SDG policy scope left the adjusted association absent, or if a validated register benchmark tied the removed component to coverage itself.

The adjusted association also shows coverage and framing to be separable dimensions, whereas a single alignment score would conflate them. This separability is the paper's core empirical contribution to the SDG-mapping literature, which has treated coverage as the primary output (Armitage, Lorenz, and Mikki, 2020; Kashnitsky et al., 2024).

The positive adjusted correlation (pattern-consistent evidence per Section 5.1) is consistent with the two dimensions being linked through topic: where one community dominates a given SDG's coverage, the framing diverges. The epistemic-communities framework (Haas, 1992) makes this reading clearer. Haas argues that "before states can agree on whether and how to deal collectively with a specific problem, they must reach some consensus about the nature and scope of the problem. . . " (Haas, 1992, p. 29). Where research dominates coverage (SDGs 3, 9), framing reflects academic norms of knowledge production (Gibbons et al., 1994, p. 3). On this reading, the coverage–semantic association reflects two communities framing the same goals differently wherever their attention diverges.

Haas cautions that consensual knowledge is often weak at explaining state behaviour (Haas, 1992, p. 30). The adjusted association links coverage to framing through topic but does not speak to policy uptake.

The directional decomposition (H1b–H1d) shows an asymmetric structure (Table 4). H1d (policy coverage) is the strongest consistent predictor, positive in 9/9 configurations. Framing divergence is wider where policy heavily covers an SDG, consistent with Haas's observation that epistemic communities occupy institutional niches (Haas, 1992, p. 30) and with the Mode 1/Mode 2 distinction (Gibbons et al., 1994).

This asymmetry is consistent with Cash et al.'s (Cash et al., 2003) observation that salience, credibility, and legitimacy are "tightly coupled" such that efforts to enhance one "normally incur a cost to the others." The SDGs where policy attention concentrates (13, 16, 17) are also those whose framings sit furthest from research norms of credibility.

5.3 The register decomposition: what the raw gap actually measures

For 16 of 17 SDGs, the register component is positive (+0.076 to +0.261, Fig. 5), meaning register uniformly inflates the apparent divergence. Studies comparing corpora that differ in register overstate semantic divergence when they use raw embedding distances. The adjusted gap alone approximates topic.

The exception is SDG 17, where the register component is negative (-0.149): register makes the two communities appear more similar than their topics warrant. Under the raw gap SDG 17 ranks near the bottom (0.177) but after register removal it jumps to the top (0.326). Policy partnership discourse uses institutional language (*countries*, *nations*, *global*) that superficially resembles research vocabulary, but the substance differs: research concerns technical terms (*model*, *method*), while policy concerns institutional coordination.

Partnership vocabulary resembles both communities' language while the coordination logic sits with policy alone, and Le Blanc (2015)'s treatment of SDG 17 fits this inversion.

SDG stratification works because the 17 goals span many topics, so a direction separating research from policy cannot also encode any one SDG topic. The method assumes topic and register can be pulled apart. Where they cannot (e.g., clinical notes vs biomedical literature), some topic would leak into the removed component.

5.4 Level 0: Institutional Commensurability

The three-level framework assumes research and policy discourse are already commensurable categories. Constructing the corpora exposes a prior condition it does not capture. The research corpus is an intersection of AI and SDG content (2,536,771 abstracts), while the curated AI-and-SDG policy family is 94 documents, and policy documents discuss AI governance and SDG implementation as separate agendas. The constraint is structural (Section 5.5): "AI and the SDGs" is not yet a settled organising category in institutional policy discourse the way it is in academic research. Level here is reference-relative: the observation applies the framework's logic one category outward, asking whether AI coverage is established within general SDG discourse on the policy side. In the formal sense of this paper's Level 1 analysis it is not a measurement (no general-SDG coverage statistic was collected), and it motivates future work.

This asymmetry maps onto the Mode 1/Mode 2 distinction (Gibbons et al., 1994) one level up. Mode 1 lets a research community define a cross-cutting category internally, while institutional discourse adopts new categories only when demanded by a concrete problem. The absence of a comparably sized "AI-and-SDG" policy corpus is therefore the gap observed before Level 1 coverage or Level 2 framing can be measured, and the study makes no claim to have filled it. The gaps measured here describe divergence within a category that barely exists on the policy side. A future Level 0 measurement (tracking co-occurrence of AI and SDG terms in institutional agendas over time) would convert this inference into a measurement.

5.5 Limitations

Semantic proximity, not substantive alignment: The study is positioned at Level 2 of a three-level construct (Section 2.3). Level 3 requires qualitative, institutional, and deliberative methods and is left to future work.

Distance metric discards distributional shape: Centroids are mean-direction measures. Under raw embeddings, distribution-aware metrics (sliced Wasserstein, Chamfer) correlated strongly with the gap ($\rho = 0.44$–0.91), but this agreement was a register artifact: they track the register component, not the adjusted semantic gap (Appendix D). After register removal the adjusted gap is likewise a mean-direction measure, and distribution-aware metrics correlate with it only weakly to moderately (e.g. Gaussian-2-Wasserstein $\rho = 0.243$, sliced Wasserstein $\rho = 0.525$). They remain an informative full-distribution complement. The covariance-shape term accounts for $\approx 79\%$ of the squared Gaussian-2-Wasserstein distance, and a replicate under the adjacent seed changes the largest per-SDG gap by at most 0.008 (Appendix D).

Unaudited upstream SDG filter: The research corpus depends on OpenAlex's native SDG-work classification to define the candidate universe before the four AI terms are applied (Section 3.3). This filter's precision and recall against ground-truth SDG relevance are not independently audited here, and the 2,536,771 retrieved abstracts reflect OpenAlex's operational rather than validated definition of SDG relevance. The corpus is large enough that composition estimates are stable, but size alone does not confirm the filter is accurate. Auditing against a manually labelled sample is left for future work.

Retrieval-method dependency: The keyword-retrieved corpus trades recall for precision. Appendix A shows concept-retrieved papers (field-of-study membership) produce closely aligned gap rankings, with a lower SDG 4 share (-7.3pp) the largest deviation between the two profiles. Re-running the full pipeline on concept-retrieved papers is the most direct path to eliminating this artefact and is left for future work. That check also reuses the keyword-corpus register-adjusted space for concept-retrieved embeddings, assuming a stable research-policy register direction. Since this assumption is untested, a native INLP re-fit is recommended before treating the track as fully independent.

Concept-retrieved corpus is unfiltered for SDG relevance: Unlike the primary keyword-retrieved corpus, which applies OpenAlex's SDG work filter before the AI terms, the concept-retrieved robustness corpus (Appendix A) restricts only to AI/ML field-of-study membership, with no SDG filter at retrieval. Its SDG labels are assigned post hoc by the trained classifier, so agreement with the keyword-retrieved corpus (Table 4) shows stability of coverage-share ranking under a different retrieval strategy, not that the concept corpus is itself SDG-relevant. Whether this unfiltered retrieval destabilises the raw-gap estimate is examined in Appendix A: the concept-retrieved configuration alone flips sign on H1a's raw column while MiniLM and SciBERT do not, a pattern consistent with the concern but not conclusive. Applying a comparable SDG filter before treating the concept corpus as a fully independent validation is recommended. Follow-up works should either apply OpenAlex's SDG filter and audit it for validation, or omit it and allow baseline non-SDG classification (Section 3.5).

Register adjustment is partial: INLP removes a linear subspace that captures the dominant corpus-level register difference, but register is multi-layered (Biber, 1988). The register interpretation rests on a structural argument (SDG stratification ensures topic cannot leak into the removed direction, Section 3.8) plus partial empirical validation against independent linguistic markers (Appendix G.2).

Per-SDG assignment is approximate: The hard argmax treats SDG discourse as separable, but the PCA space (Figure 4) shows real-world SDG discourse is fuzzy. Leave-one-out of SDG 4 (audited in Appendix B.1) preserves the adjusted association ($\rho = 0.597$, $p = 0.016$). Future work with multi-class SDG classification is recommended.

Cross-sectional design: The study observes one period (2018–2025) against a similarly timed policy corpus. It cannot assess whether alignment is improving or deteriorating, or whether specific events (e.g., the EU AI Act 2024, IPCC AR6 2022) (European Union, 2024; IPCC, 2022) shifted research framing.

5.6 Contribution and Implications

The primary contribution of this paper is the measurement framework. The AI–SDG findings are a first application to be checked against a larger, more diverse policy corpus: the cancellation, the SDG 17 inversion, and the positive adjusted association are properties of this specific pair of corpora at this moment in time. The framework advances AI–SDG mapping from Level 1 (lexical correspondence) to Level 2 (observed semantic proximity), with explicit marking of Level 3 boundaries (Section 2.3). The cross-sensitivity table (Table 17) is a reusable template that shows how stable every ranking position is across measurement choices. The SDG-stratified INLP adaptation (training the corpus classifier on balanced topic–corpus strata so that no linear direction can encode both topic and register) applies to any cross-corpus comparison where topic and register are confounded, and the protocol is portable to any research-policy interface that shares a category system and embeds both sides in a common semantic space.

Beyond the measurement advance, the results carry a direct implication for how SDG engagement is read. Coverage and framing are separable, correlating at only $\rho = 0.544$, confirming the premise that shared coverage does not imply shared framing. Coverage, the Level 1 metric existing studies report, captures only a weak form of salience and leaves credibility and legitimacy aside (Cash et al., 2003). The adjusted gap is a Level 2 metric that begins to measure framing compatibility. The high-gap SDGs (17, 3, 16, 9, 13) are candidates for boundary-organising practices (Guston, 2001): policy synthesis briefs, joint workshops, or funder-required summaries. Cash et al. (Cash et al., 2003) caution that translation alone cannot resolve the tradeoffs among salience, credibility, and legitimacy.

6 CONCLUSION

The Introduction asked how academic AI-for-sustainability research and institutional policy discourse differ in SDG coverage and semantic framing, and how these two dimensions are related. This paper demonstrated that coverage divergence (which SDGs each community prioritises) and within-SDG semantic framing divergence (how differently they discuss the same goal) are separable dimensions that can diverge independently. The linkage is asymmetric: policy coverage is the strongest directional predictor of framing divergence (positive in 9/9 configurations), while research coverage shows no consistent signal. Where policy attention concentrates, the discourse resembles Mode 2 knowledge production (Gibbons et al., 1994), and its framings sit further from research norms. Both regularities fit the account of two distinct knowledge cultures addressing the same goals (Caplan, 1979; Cash et al., 2003).

The takeaway for practice is that an SDG label does not by itself evidence alignment: a tag records that a text is about a goal without recording how it frames that goal. Because funders and policymakers read SDG counts as proof of engagement (Heilinger, Kempt, and Nagel, 2024; Heras-Saizarbitoria, Urbieta, and Boiral, 2022), interpreting that engagement correctly requires the two-dimensional reading developed here, coverage alongside framing with the register component made explicit.

The asymmetric corpus construction is itself suggestive: the difficulty of building a symmetric AI-SDG policy corpus may reflect that "AI and the SDGs" is not yet a settled organising category in policy discourse the way it already is in research (Section 5.4). Level 3, whether the divergences documented here reflect productive exploration, institutional failure, or different knowledge cultures pursuing different mandates, requires qualitative, institutional, and deliberative methods that lie beyond the scope of embedding analysis.

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A CONCEPT-BASED RESEARCH RETRIEVAL

This section reports results of using the concept-retrieved research corpus. This test uses field-of-study membership for the same two AI fields, mirroring the OECD.AI definition (OECD, 2024), and re-scored the resulting corpus through the identical pipeline (segmentation, Section 3.1; classifier, Section 3.5). No SDG filter was applied at retrieval. See Section 5.5 for this limitation. Concept retrieval used OpenAlex concept tags for Artificial Intelligence and Machine Learning, capped at 100,000 papers total. This scale sits within the stable range of the sample-stability ladder reported in Appendix E (coverage gap 0.056 and mean within-SDG semantic gap (raw) 0.342, consistent with the 2,000,000-paper estimates of 0.056 and 0.340) and exceeds every corpus size used in prior SDG classification and mapping benchmarks: the OSDG community dataset (30,532 texts) (Pukelis et al., 2022), the SDG benchmark encoder evaluation corpus (35,001 excerpts) (Gjorgjevikj et al., 2025), the SDGi corpus (20,267 segments) (SDGi, 2024), the Aurora expert-validated set (4,971 texts) (Vanderfeesten, Spielberg, and Gunes, 2020), and the Kashnitsky et al. (2024) benchmark (6,000 papers)—while remaining far smaller than the full research corpus of 2,536,771 abstracts, making it a computationally tractable substrate for OpenAlex concept tagging.

Figure 7 compares SDG coverage shares between the keyword-retrieved and concept-retrieved corpora. The two profiles are closely aligned (Spearman $\rho = 0.948$, Kendall $\tau = 0.868$). The largest deviations are SDG 4 (-7.3pp, the known ML-vocabulary artefact, Appendix B.1) and SDG 9 ($+5.8$pp). Within-SDG coverage-gap and semantic-gap rankings are also stable against the keyword baseline (coverage-gap rank $\rho = 0.868$; semantic-gap rank $\rho = 0.814$, Table 17). The main findings are therefore reported as robust to query-term choice, while both corpora remain conditional on their respective retrieval strategies.

The unfiltered-retrieval concern also manifests in the interaction grid (Table 4). On H1a's raw semantic-gap column, the concept-retrieved configuration is the only one whose sign flips against the rest of the encoder–classifier grid (Concept LR: +0.108, versus negative values in the other eight configurations), while the MiniLM and SciBERT tracks — also different encoders — show no comparable sign flip. (H1c's raw column is mixed across all configs, so no single track uniquely flips there.) This does not prove the unfiltered-retrieval concern causes the instability, but it is the pattern that concern would predict. Time constraints prevented applying a comparable SDG filter to this alternative corpus; applying one is recommended before treating it as a fully independent validation.

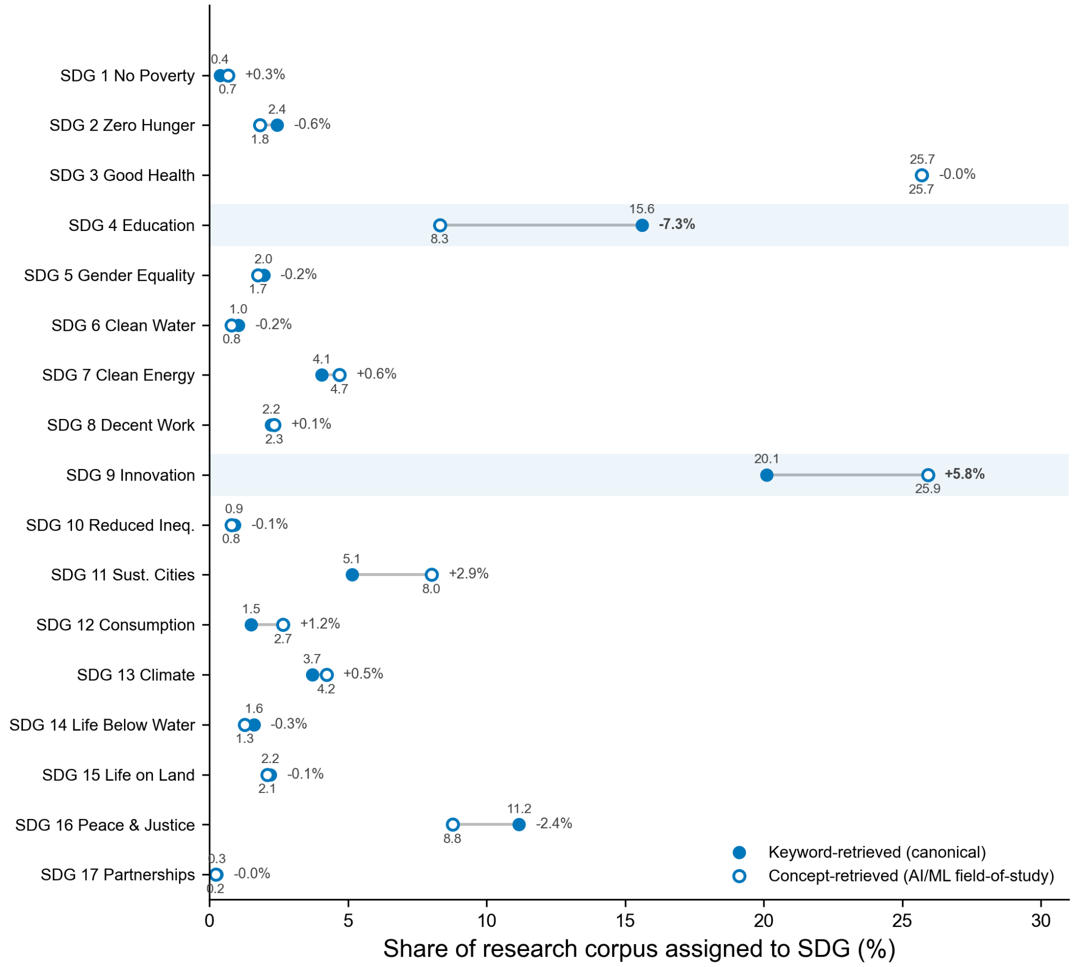

Figure 7. SDG coverage-share comparison: keyword-retrieved versus concept-retrieved (AI/ML field-of-study) research corpus.

Notes: Δ is concept minus keyword share (

Table 5 reproduces the per-SDG reference values (Section 4.4) for the concept-retrieved corpus: the semantic-gap family (raw, adjusted, register component) and the coverage-predictor family (coverage gap, signed dominance, research share, policy share). Signed dominance = research share − policy share (pp); coverage gap = |research − policy| (pp). These are the per-SDG inputs behind the Concept rows in the interaction grid (Table 4) and the cross-sensitivity tables.

Table 5. Concept-retrieved corpus — per-SDG reference values (semantic-gap and coverage-predictor families).

SDG	Description	Semantic gaps			Coverage predictors			
		Raw	Adj.	Reg.	Cov.	SDom	Res %	Pol %
1	No Poverty	0.827	0.440	+0.387	3.9	-3.9	0.7	4.5
2	Zero Hunger	0.428	0.137	+0.291	2.5	-2.5	1.8	4.3
3	Good Health	0.663	0.454	+0.209	21.0	+21.0	25.7	4.7
4	Education	0.402	0.224	+0.178	3.5	+3.5	8.3	4.8
5	Gender Equality	0.575	0.333	+0.241	3.3	-3.3	1.7	5.1
6	Clean Water	0.384	0.180	+0.204	2.9	-2.9	0.8	3.7
7	Clean Energy	0.514	0.242	+0.272	0.4	+0.4	4.7	4.3
8	Decent Work	0.625	0.363	+0.262	2.0	-2.0	2.3	4.3
9	Innovation	0.421	0.330	+0.092	21.6	+21.6	25.9	4.3
10	Reduced Ineq.	0.458	0.312	+0.146	2.1	-2.1	0.8	2.9
11	Sust. Cities	0.693	0.458	+0.236	2.7	+2.7	8.0	5.3
12	Consumption	0.400	0.280	+0.120	0.9	-0.9	2.7	3.5
13	Climate	0.598	0.266	+0.332	7.1	-7.1	4.2	11.3
14	Life Below Water	0.504	0.253	+0.251	2.5	-2.5	1.3	3.8
15	Life on Land	0.346	0.139	+0.207	2.1	-2.1	2.1	4.2
16	Peace & Justice	0.541	0.327	+0.214	8.5	-8.5	8.8	17.3
17	Partnerships	0.312	0.521	-0.210	11.5	-11.5	0.2	11.7
Mean	—	0.511	0.309	+0.202	5.8	+0.0	5.9	5.9

Notes: The semantic-gap and coverage-predictor structure is retained. Signed dom. = research % − policy % (pp); Reg. Comp. = raw − adjusted. The concept-retrieved corpus reuses the MPNet register-adjusted space (Section 3.8).

B DIAGNOSTICS FOR REFERENCE CLASSIFIER

B.1 SDG 4 Lexical Artefact Audit: The ML 'Learning' Overlap

SDG 4 is the only goal audited because it is the only one carrying an a priori lexical-overlap concern: "learning" is shared ML and education vocabulary, and prior AI–SDG inventories place SDG 4 outside the leading tier of SDG 3 and SDG 7 (Singh et al., 2024; Cowls et al., 2021; Nedungadi et al., 2024). In Singh et al. (2024)'s 20,511-document corpus, SDG 4 ranks third of 17 but with only 2,312 publications, roughly a third of the counts for SDG 3 (6,829) and SDG 7 (6,683); its third-place research share here is suspect on independent grounds. The audit does not relabel records. It tests whether the concern is empirically plausible.

The keyword groups are defined as follows. The ML group comprises 14 terms: *machine learning, deep learning, reinforcement learning, supervised learning, unsupervised learning, neural network, model, models, train, trained, training, algorithm, classification, prediction*. The education group comprises 17 terms: *education, educational, school, schools, student, students, teacher, teachers, classroom, curriculum, learning outcome, pedagogy, university, universities, course, courses, teaching*. Note that bare "learning" is not a matched term on either side. It appears only within compound phrases (*machine learning* on the ML side, *learning outcome* on the education side), reconciling the subsection title with the actual matching behaviour. Matching is case-insensitive, whole-word (`\b...\b` regex), and unstemmed, applied to the segmented title+abstract research text. A record is classified as "both" if it contains at least one term from each group; otherwise it is ML-only, education-only, or neither. The machine-readable keyword lists and aggregate counts are stored in `4_outputs/appendix/mpnet/a3_sdg4_audit/data/sdg4_lexical_audit_summary.json`.

Two methodological caveats apply. First, the audit operates at the segment level (3,105,144 rows, one per text segment), not the paper level. Roughly 34% of multi-segment papers carry differing labels across their segments, so segment-level percentages are segment-weighted and may overrepresent long abstracts. The matched non-SDG 4 sample is a seed-42 reservoir draw from non-SDG-4 segment rows. Second, the ML list intentionally includes generic method vocabulary (*model, training, algorithm*) that is near-universal in an AI corpus. The diagnostic signal is therefore not the ML-presence rate itself but the education-versus-ML asymmetry across subsets (44.4% education-only in SDG 4 versus 4.0% and 2.5% in the comparison sets).

Table 6 compares the lexical composition of SDG 4-assigned research with a matched non-SDG 4 sample and with SDG 9, using these two keyword groups. The result is a negative test of the false-positive hypothesis: education-specific vocabulary alone covers 44.4% of SDG 4-assigned records and overlaps ML vocabulary in a further 23.8%, while only 11.8% are ML-only. The comparison sets are inverted, with ML-only vocabulary dominating the non-SDG 4 sample and the SDG 9 set (50.3% and 61.5%). The SDG 4 signal is therefore not the same technical language that drives SDG 9. But the 23.8% "both" share (versus 2.6% and 2.9% elsewhere) shows the two vocabularies do overlap materially within the SDG 4-assigned set.

The main analysis therefore retains the canonical LR argmax assignment but continues to treat SDG 4 as artefact-affected. The most accurate reading is a learning-related assignment whose interpretation is blurred by lexical overlap: neither "clean education" nor "pure ML artefact."

Table 6. SDG 4 lexical artefact audit.

Corpus subset	N	ML-only %	Education-only %	Both %	Neither / unclear %
SDG 4-assigned research	470002	11.8	44.4	23.8	19.9
Non-SDG4 research sample	470002	50.3	4.0	2.6	43.1
SDG 9-assigned research	577833	61.5	2.5	2.9	33.1

Notes: The audit only checks whether SDG 4-assigned research is disproportionately dominated by machine-learning vocabulary, education vocabulary, or both.

B.2 Centroid Similarity

This subsection reports the pairwise cosine-similarity heatmap (Figure 8) for the 17 canonical SDG centroids. The centroids are the per-SDG mean training embeddings (L2-normalised) used by the LR classifier.

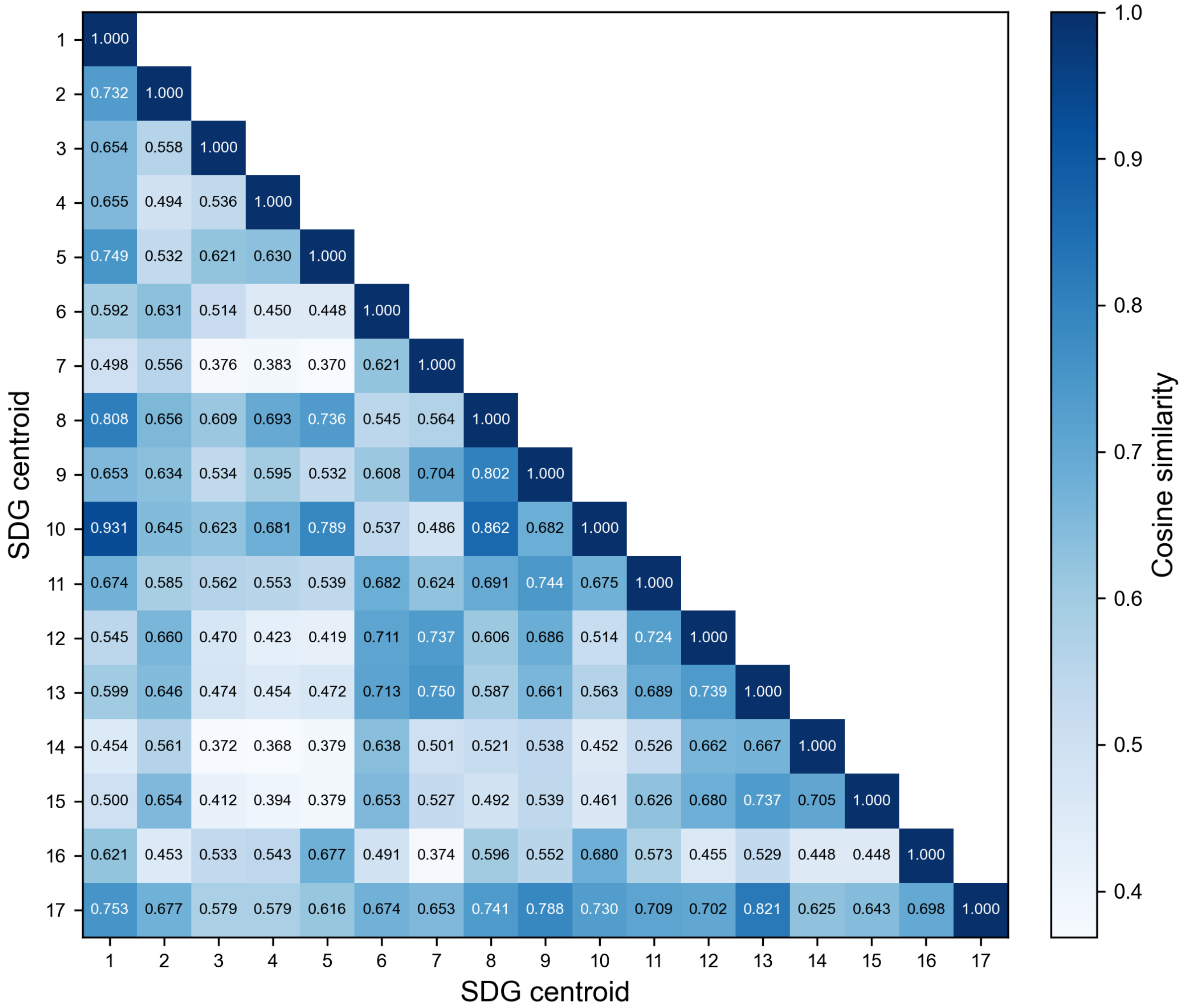

Figure 8. Pairwise cosine similarity between canonical SDG centroids (lower triangle).

Figure 8 shows the full 17×17 similarity grid. Reading the grid, three observations stand out. First, the diagonal dominates: most centroids are far more similar to themselves than to any neighbour, which is why the supervised LR classifier (Table 3) can build on this well-separated base to reach 0.816. Second, a small set of off-diagonal cells is elevated: SDG 11's centroid neighbours SDG 9 (cosine 0.744), SDG 8 lies near SDGs 1 and 10 (cosine 0.808 and 0.862), and SDGs 1 and 10 are themselves the closest off-diagonal pair (cosine 0.931) – these are the proximities that make the SDG 10 and SDG 8 scores in Section 4.1 indicative rather than exact. Third, SDG 17 is the outlier: its row and column are the flattest, and its nearest neighbour is SDG 13 (cosine 0.821), reflecting the overlap between climate and partnership discourse noted in Section 4.1.

C LEXICAL ILLUSTRATION OF THE SEMANTIC GAP

Table 7 lists the ten TF-IDF-weighted unigrams and bigrams most distinctive of the research and policy sides for each SDG, computed on deterministic samples of texts assigned to each SDG. Two groups of stopwords are excluded because they reflect the retrieval strategy (Section A) rather than SDG-specific framing: generic ML-methodology vocabulary (*model*, *models*, *learning*, *neural*, *deep*, *machine*, *algorithm*, *algorithms*, *method*, *methods*, *proposed*, *network*, *networks*, *accuracy*, *performance*, *training*, *classification*, *feature*, *features*, *layer*, *layers*, *dataset*, *datasets*, *analysis*, *prediction*, *findings*, *factors*, *regression*, *based*, *effect*, *variables*, *used*, *associated*, *different*) and generic structural terms (*abstract*, *article*, *chapter*, *data*, *development*, *figure*, *goal*, *goals*, *paper*, *research*, *result*, *results*, *sdg*, *sdgs*, *study*, *sustainable*, *table*, *text*, *using*). No further analysis is performed on these terms. The table is included for illustrative reference only.

Table 7. Distinctive TF-IDF terms for all 17 SDGs.

SDG	Gap	Research-side distinctive terms	Policy-side distinctive terms
1	0.415	alleviation, china, approach, zakat, satellite, household, studies, indonesia, literature, relationship	social, cent, national, people, protection, services, population, vulnerable, living, million
2	0.299	crop, detection, plant, diseases, images, leaf, precision, techniques, soil, yield	population, food, children, prevalence, worst, access, hunger, million, births, ratio
3	0.494	patients, cancer, cells, images, clinical, detection, disease, potential, genes, predict	health, services, care, national, population, births, mortality, rate, people, access
4	0.267	cognitive, brain, neurons, sleep, agricultural, food, cells, motor, memory, poor	education, school, secondary, primary, national, quality, children, percent, access, vocational
5	0.278	menstrual, poor, pain, mothers, differences, relationship, patients, fertility, stress, sex	gender, violence, equality, national, rights, girls, men, cent, sexual, domestic
6	0.292	irrigation, soil, agricultural, crop, salinity, agriculture, sup, potential, sub, yield	sanitation, access, drinking, supply, national, wastewater, cent, services, population, clean
7	0.428	agricultural, poverty, agriculture, battery, optimization, approach, load, drying, poor, artificial	access, electricity, renewable, clean, worst best, population, affordable, index worst, cent, national
8	0.387	detection, image, food, recognition, leaf, plant, cnn, convolutional, disease, techniques	growth, employment, gdp, worst, labour, best, economic, cent, government, tourism
9	0.314	agricultural, agriculture, iot, detection, food, crop, plant, farming, farmers, soil	infrastructure, innovation, national, government, public, countries, strategy, access, population, intelligence
10	0.271	wealth, income inequality, effects, evidence, studies, que, distribution, literature, mobility, poor	people, rights, countries, migration, national, cent, disabilities, persons, discrimination, target
11	0.449	detection, image, accident, traffic, object, poor, recognition, spatial, fusion, agricultural	housing, city, urban, transport, cities, public, waste, national, plan, green
12	0.296	food, agricultural, detection, quality, potential, behavior, samples, organic, design, compounds	waste, consumption, national, recycling, production, procurement, management, public, economy, circular
13	0.480	soil, agricultural, crop, moisture, drought, sub, rice, stress, agriculture, yield	climate, change, emissions, countries, national, action, global, energy, nations, united
14	0.358	detection, images, aquaculture, food, bacterial, shrimp, cells, imaging, farming, techniques	marine, fish caught, biodiversity, rate, ocean, worst, population, sea, embodied, coastal
15	0.239	soil, crop, agricultural, plant, spectral, images, sensing, detection, remote, field	biodiversity, forest, worst, national, protected, population, area, freshwater, species, best
16	0.329	poor, food, covid-, pain, image, detection, social, studies, patients, brain	international, nations, peace, rights, united, security, human, law, world, global
17	0.177	sentiment, tweets, covid-, twitter, social, agricultural, dan, yang, zakat, media	countries, oecd, international, nations, united, agenda, report, financing, global, challenges

Notes: Terms are TF-IDF-weighted unigrams and bigrams from deterministic samples of research and policy texts assigned to each SDG.

D DISTANCE-FUNCTIONAL ROBUSTNESS TABLE

Table 8 reports the per-SDG research–policy gap under the full battery of distribution-aware metrics, together with the primary semantic gap. Part 1 covers the full-corpus exact metrics (sliced Wasserstein (Bonneel et al., 2015), Chamfer (Borgefors, 1988), Hausdorff, Gaussian-2-Wasserstein (Villani, 2008), Grassmann, linear MMD (Sejdinovic et al., 2013)); part 2 covers the sampled metrics (exact EMD (Rubner, Tomasi, and Guibas, 2000; Kusner et al., 2015), Sinkhorn (Cuturi, 2013), energy (Székely and Rizzo, 2013), squared RBF MMD, C2ST AUC) plus the sampled semantic gap. The final row of each part reports Spearman ρ between each metric's ranking and the primary (adjusted) ranking.

Why the raw robustness result does not carry over: Under raw embeddings the distribution-aware metrics ($\rho = 0.44$–0.91) tracked the *register* component, not topic: register has larger variance than the raw gap (ratio 1.210) and correlates $\rho = 0.748$ with it. After INLP removes register, the metrics track topic-space shape – most correlate moderately with the adjusted gap (sliced Wasserstein $\rho = 0.525$, exact EMD $\rho = 0.694$), Gaussian-2-Wasserstein/squared MMD weaker ($\rho \approx 0.24$–0.26), Sinkhorn/C2ST/Grassmann near-zero or negative ($\rho = 0.047$, -0.164, -0.243; elevating SDGs 12 and 1 marginally). The shape term is $\approx 79\%$ of the Gaussian-2-Wasserstein distance; a seed-replicate shifts the largest gap by ≤ 0.008.

Table 10 re-runs H1a–H1d with distribution-aware topic gaps. The positive coverage–framing link replicates: sliced Wasserstein ($\rho = 0.554$, $p = 0.022$), linear MMD ($\rho = 0.603$, $p = 0.012$), and energy ($\rho = 0.537$, $p = 0.027$) match the centroid estimate ($\rho = 0.544$, $p = 0.025$); squared RBF MMD directionally consistent. The distributional view adds one nuance the centroid misses: H1c (research coverage) turns positive under sliced-Wasserstein/energy/RBF-MMD despite being null on the centroid gap. Metrics with weak/negative rank agreement (Sinkhorn, C2ST, Grassmann) are null too, except Sinkhorn's H1d ($\rho = -0.505$, $p = 0.041$).

Table 8. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 1 of 2: full-corpus exact metrics).

SDG	Canonical	SWD	Chamfer	Hausdorff	Gauss. W_2	Grassmann	linear MMD
1	0.191	0.009	0.509	0.600	0.684	3.727	0.067
2	0.089	0.008	0.388	0.454	0.618	3.629	0.057
3	0.319	0.011	0.454	0.581	0.717	3.717	0.129
4	0.158	0.010	0.399	0.487	0.607	3.307	0.096
5	0.151	0.010	0.429	0.529	0.646	3.514	0.095
6	0.130	0.009	0.417	0.475	0.639	3.598	0.080
7	0.170	0.010	0.427	0.522	0.687	3.729	0.105
8	0.184	0.008	0.474	0.564	0.658	3.566	0.065
9	0.228	0.010	0.430	0.556	0.671	3.444	0.104
10	0.195	0.008	0.456	0.517	0.617	3.343	0.062
11	0.209	0.009	0.446	0.542	0.677	3.484	0.091
12	0.156	0.009	0.467	0.566	0.677	3.462	0.094
13	0.219	0.010	0.417	0.498	0.667	3.667	0.117
14	0.132	0.010	0.415	0.510	0.675	3.753	0.098
15	0.084	0.008	0.394	0.464	0.578	3.421	0.056
16	0.242	0.010	0.417	0.520	0.640	3.333	0.103
17	0.326	0.009	0.461	0.490	0.595	2.993	0.104
ρ vs. canonical	–	0.525	0.485	0.444	0.243	-0.243	0.620

Notes: SWD = sliced Wasserstein; MMD^2 = squared RBF maximum mean discrepancy; Gauss. W_2 = Gaussian 2-Wasserstein; C2ST = classifier two-sample test AUC. The final row reports Spearman ρ between each metric's ranking and the canonical ranking across the 17 SDGs.

Table 9. Per-SDG research–policy gap under distribution-aware metrics versus the canonical centroid gap (Part 2 of 2: sampled metrics; continuation of Table 8).

SDG	Canonical	EMD	Sinkhorn	Energy	MMD^2	C2ST AUC	Centroid (samp.)
1	0.191	0.645	0.623	0.056	0.025	0.969	0.186
2	0.089	0.533	0.516	0.054	0.026	0.979	0.088
3	0.319	0.665	0.597	0.105	0.046	0.985	0.315
4	0.158	0.554	0.523	0.084	0.040	0.988	0.157
5	0.151	0.581	0.543	0.082	0.038	0.984	0.152
6	0.130	0.546	0.547	0.073	0.036	0.986	0.128
7	0.170	0.581	0.545	0.094	0.045	0.990	0.170
8	0.184	0.620	0.596	0.054	0.024	0.978	0.185
9	0.228	0.625	0.560	0.090	0.040	0.984	0.228
10	0.195	0.602	0.636	0.051	0.023	0.964	0.191
11	0.209	0.606	0.580	0.076	0.035	0.977	0.213
12	0.156	0.605	0.605	0.081	0.037	0.987	0.154
13	0.219	0.577	0.485	0.102	0.048	0.983	0.217
14	0.132	0.555	0.538	0.091	0.044	0.984	0.134
15	0.084	0.515	0.515	0.053	0.026	0.982	0.083
16	0.242	0.604	0.495	0.088	0.039	0.983	0.241
17	0.326	0.582	0.499	0.083	0.037	0.980	0.319
ρ vs. canonical	–	0.694	0.047	0.444	0.260	-0.164	1.000

Notes: EMD = exact 1-Wasserstein; MMD^2 = squared RBF-MMD; Centroid (samp.) = sampled centroid-gap control column.

Table 10. H1a–H1d coverage-predictor vs semantic-gap Spearman correlations under distribution-aware semantic gaps (adjusted embeddings). Distributional gaps replace the centroid gap as the outcome vector; predictors and test are identical to Table 4.

	H1a Cov. gap	H1b Dominance	H1c Research cov.	H1d Policy cov.
Sliced Wasserstein	+0.554*	+0.061	+0.539*	+0.397
Linear MMD	+0.603*	+0.022	+0.466†	+0.439†
Energy distance	+0.537*	+0.074	+0.532*	+0.384
Squared RBF MMD	+0.473†	+0.113	+0.551*	+0.296
Exact EMD	+0.304	+0.147	+0.167	+0.217
Chamfer	+0.103	-0.115	-0.350	-0.006
Modified Hausdorff	+0.159	+0.152	+0.127	+0.066
Gaussian-2-Wasserstein	-0.007	+0.240	+0.262	-0.042
Sinkhorn	-0.199	+0.397	-0.176	-0.505*
C2ST AUC	+0.270	+0.275	+0.375	-0.139
Grassmann	-0.186	+0.034	+0.012	-0.228

Notes: Spearman ρ with two-sided Monte Carlo permutation p-values (100,000 resamples, seed 42), $n = 17$ SDGs; $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{\dagger}p < 0.10$. Excluding SDG 4 ($n = 16$) leaves every conclusion unchanged; the full $n = 16$ grid is in the analysis JSON (distributional_h1_interaction.json).

E SAMPLE-STABILITY ROBUSTNESS CHECK

To test whether the headline findings depend heavily on research-corpus size, this study reran the downstream research-side analysis on repeated random subsamples drawn from the pre-embedded, pre-scored OpenAlex corpus. For each of 11 sampled tiers (from 1,000 to 2,000,000 papers), 100 independent samples were drawn without replacement, the research coverage profile was recomputed, sampled research centroids were rebuilt, the coverage gap against the fixed policy profile was recalculated, and the within-SDG semantic gap was recomputed on both the raw and the register-adjusted (topic-only) embeddings. Policy-side quantities were held fixed.

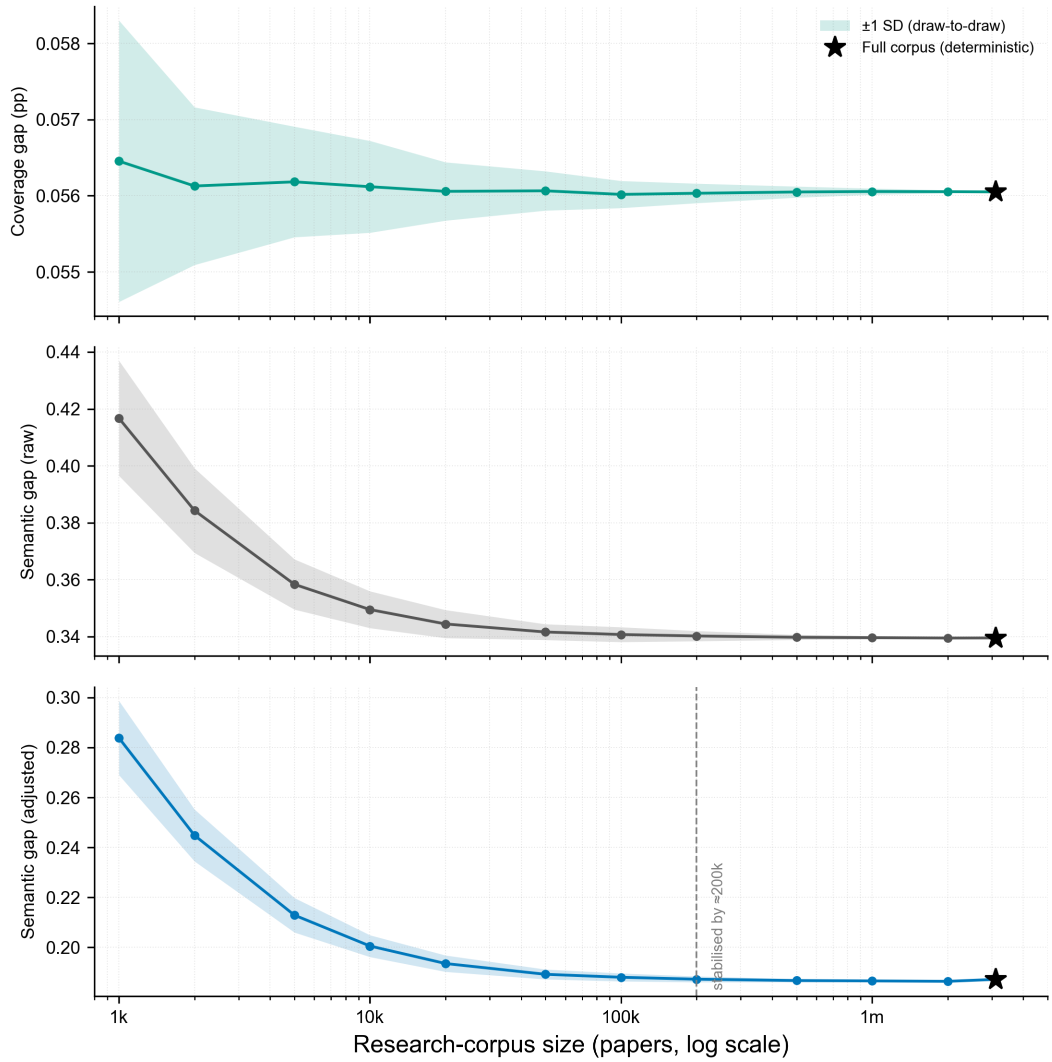

Figure 9. Sample-stability ladder for the research corpus.

Notes: Coverage gap is the unweighted mean absolute deviation of the research SDG coverage shares from the fixed policy SDG coverage shares. Lines show the per-tier mean and shaded bands the draw-to-draw ± 1 SD of the within-SDG semantic gap on the raw and register-adjusted embeddings (100 draws per tier); the star marks the deterministic full-corpus value, which has no dispersion term.

Plain-language guide. The stability ladder tracks three metrics: (1) coverage gap, how far the research SDG coverage shares sit from the fixed policy SDG coverage shares; (2) semantic gap (raw), how different research and policy language are within the same SDG before register removal; (3) semantic gap (adjusted), the same gap after the register direction is projected out, isolating topic. The coverage gap is stable at $\sim$0.056 across every sampled tier. Both the raw semantic gap and the register-adjusted semantic gap stabilise to within 0.001 of their full-corpus values (raw $\sim$0.340; adjusted $\sim$0.187) by roughly 200,000 papers (raw $\sim$0.340; adjusted $\sim$0.187). From 500,000 papers upward, draw-to-draw dispersion is negligible. The full 3,105,144-segment result is retained as the primary estimate because it uses all available data, yields the tightest values, and enables per-SDG subgroup detail and rank-order precision that the plateau minimum does not guarantee; the substantive conclusions are not artefacts of sample size choice.

Balanced-subset rank stability: The ladder holds the policy side fixed while the research side grows, so the two sides are never size-matched within a tier. Because research (2,536,771 abstracts) outweighs policy (6,367 documents) by about 398:1, this study also tested whether the within-SDG semantic-gap *ranking* survives when the research side is subsampled to a size comparable to the policy corpus. Across 100 independent draws at the 50,000-paper tier, the gap ranking reproduces the full-corpus ordering at Spearman $\rho = 0.986$ (SD 0.009), already $\rho = 0.936$ at 10,000 papers and $\rho = 0.990$ at 100,000 papers, converging to $\rho = 1.000$ at the full corpus.

F MODEL SELECTION: GRID SEARCH PROTOCOL AND RATIONALE

This appendix supplies the commentary on the MLP robustness and zero-shot nearest-centroid results reported without interpretation in Section 4.5 (and the held-out test results of Section 4.1), and reproduces the full 32-candidate ranking behind the champion selection.

The MLP test macro-F1 (0.826) is a single held-out evaluation of the champion MLP retrained on the full training pool (n=52,835), distinct from the 5-fold CV mean of 0.8192 $\pm$ 0.0032 in Section 4.1.

Held-out test performance quantifies the cost of each simplification. The champion MLP reaches macro-F1 0.826 and micro-F1 0.8396, only 1.0 and 0.9 points above LR (0.816 / 0.831) with a near-identical per-SDG ordering (Spearman $\rho = 0.980$), so its extra capacity yields negligible accuracy and the interpretability-first choice of LR is justified. The zero-shot nearest-centroid baseline is weaker – macro-F1 0.760 and micro-F1 0.7743, 5.6 points below LR – because it has no trained decision boundary, only the unweighted mean of reference embeddings, and the penalty is concentrated on the goals hardest to separate for all methods (SDG 10: 0.485 vs 0.610; SDG 17: 0.671). ZS still recovers the same ranking ($\rho = 0.930$ vs LR) and far exceeds the 1/17 random baseline, so it remains a usable unsupervised sanity check rather than a competitive classifier.

Table 11. Model selection ranking (full): 5-fold GroupKFold cross-validated macro-F1 across all 32 candidates (16 MLP + 16 LR).

Rank	Config	CV macro-F1
1	MLP 3L/256h, lr=0.0003 (best)	0.8192 $\pm$ 0.0032
2	MLP 2L/256h, lr=0.0003	0.8184 $\pm$ 0.0030
3	MLP 6L/384h, lr=0.0003	0.8183 $\pm$ 0.0023
4	MLP 2L/384h, lr=0.0003	0.8183 $\pm$ 0.0022
5	MLP 4L/256h, lr=0.001	0.8181 $\pm$ 0.0017
6	MLP 2L/256h, lr=0.001	0.8180 $\pm$ 0.0038
7	MLP 3L/384h, lr=0.0003	0.8176 $\pm$ 0.0015
8	MLP 4L/384h, lr=0.0003	0.8175 $\pm$ 0.0017
9	MLP 4L/256h, lr=0.0003	0.8169 $\pm$ 0.0024
10	MLP 6L/256h, lr=0.0003	0.8168 $\pm$ 0.0026
11	MLP 3L/256h, lr=0.001	0.8165 $\pm$ 0.0030
12	MLP 2L/384h, lr=0.001	0.8162 $\pm$ 0.0053
13	MLP 6L/384h, lr=0.001	0.8156 $\pm$ 0.0039
14	MLP 6L/256h, lr=0.001	0.8153 $\pm$ 0.0027
15	MLP 4L/384h, lr=0.001	0.8151 $\pm$ 0.0024
16	MLP 3L/384h, lr=0.001	0.8151 $\pm$ 0.0015
17	LR C=3.0, cw=None (chosen)	0.8101 $\pm$ 0.0020
18	LR C=2.0, cw=None	0.8098 $\pm$ 0.0018
19	LR C=5.0, cw=None	0.8097 $\pm$ 0.0017
20	LR C=1.0, cw=None	0.8087 $\pm$ 0.0025
21	LR C=3.0, cw=balanced	0.8079 $\pm$ 0.0014
22	LR C=2.0, cw=balanced	0.8077 $\pm$ 0.0016
23	LR C=10.0, cw=None	0.8075 $\pm$ 0.0020
24	LR C=5.0, cw=balanced	0.8073 $\pm$ 0.0017
25	LR C=1.0, cw=balanced	0.8069 $\pm$ 0.0026
26	LR C=10.0, cw=balanced	0.8063 $\pm$ 0.0013
27	LR C=30.0, cw=None	0.8039 $\pm$ 0.0023
28	LR C=30.0, cw=balanced	0.8022 $\pm$ 0.0016
29	LR C=100.0, cw=None	0.7967 $\pm$ 0.0010
30	LR C=100.0, cw=balanced	0.7942 $\pm$ 0.0017
31	LR C=0.1, cw=balanced	0.7919 $\pm$ 0.0032
32	LR C=0.1, cw=None	0.7904 $\pm$ 0.0032

G REGISTER REMOVAL: CONVERGENCE AND VALIDATION

G.1 Convergence

The full INLP procedure and identification argument are given in Methodology (Section 3.8). This appendix reports the convergence behaviour of the procedure and the construct validation of the component it removes. The adjusted results are the primary decomposition, not sensitivity diagnostics.

Sampling design. Two details of the sampling in Section 3.8 bear on how the adjusted gap should be read. First, n_{target} is one global constant – the smallest available count across all SDG–corpus pools, 1123 segments for MPNet – so every SDG contributes the same number of research and policy texts at every iteration regardless of how large its actual pool is; how much text an SDG has cannot change how much register adjustment it receives. Second, the stratified sample is redrawn independently at each iteration (seeded per iteration, without replacement within each draw), so the learned directions do not hinge on any single subsample of the pool. One residue is unavoidable: for the pool that binds the global minimum, all available texts are used at every iteration, so that one stratum sees identical samples throughout; this affects only the smallest pool, treats both corpora identically, and cannot differentiate well-covered from sparsely-covered goals.

Convergence results. INLP removal converges fast. Held-out accuracy falls from 0.978 at iteration 1 to 0.498 by iteration 62, and the mean gap shrinks by 41% before flattening. Almost all of that happens in the first 15 directions: the gap's correlation with the raw (un-removed) gap drops from 0.995 to about 0.352 and stays there. Figure 10 shows the per-iteration paths for all three encoders.

The decomposition identifies the SDG 17 inversion discussed in Results (Section 4.3, Fig. 5): the register component is positive for all SDGs except SDG 17 (range: +0.076 to +0.261, smallest SDG 10, largest SDG 13), the only negative value, where register *increases* the apparent research–policy similarity of the partnership goal.

Cross-config replication. The same iterative procedure was run on the MiniLM and SciBERT encoder tracks (the latter two on the S1 subset, by design). All three reach the ≤ 0.5 stopping threshold. Held-out register-classification accuracy falls from 0.965–0.978 at iteration 1 to 0.491–0.498 at convergence, and all three compress sharply in the first 15 iterations (to 0.539–0.685) before plateauing at the majority-class baseline. The iteration count required to reach the threshold differs across encoders, but the qualitative convergence — a rapid early drop followed by a plateau — is identical, confirming that the convergence behaviour is a property of the INLP procedure rather than of the MPNet embedding space.

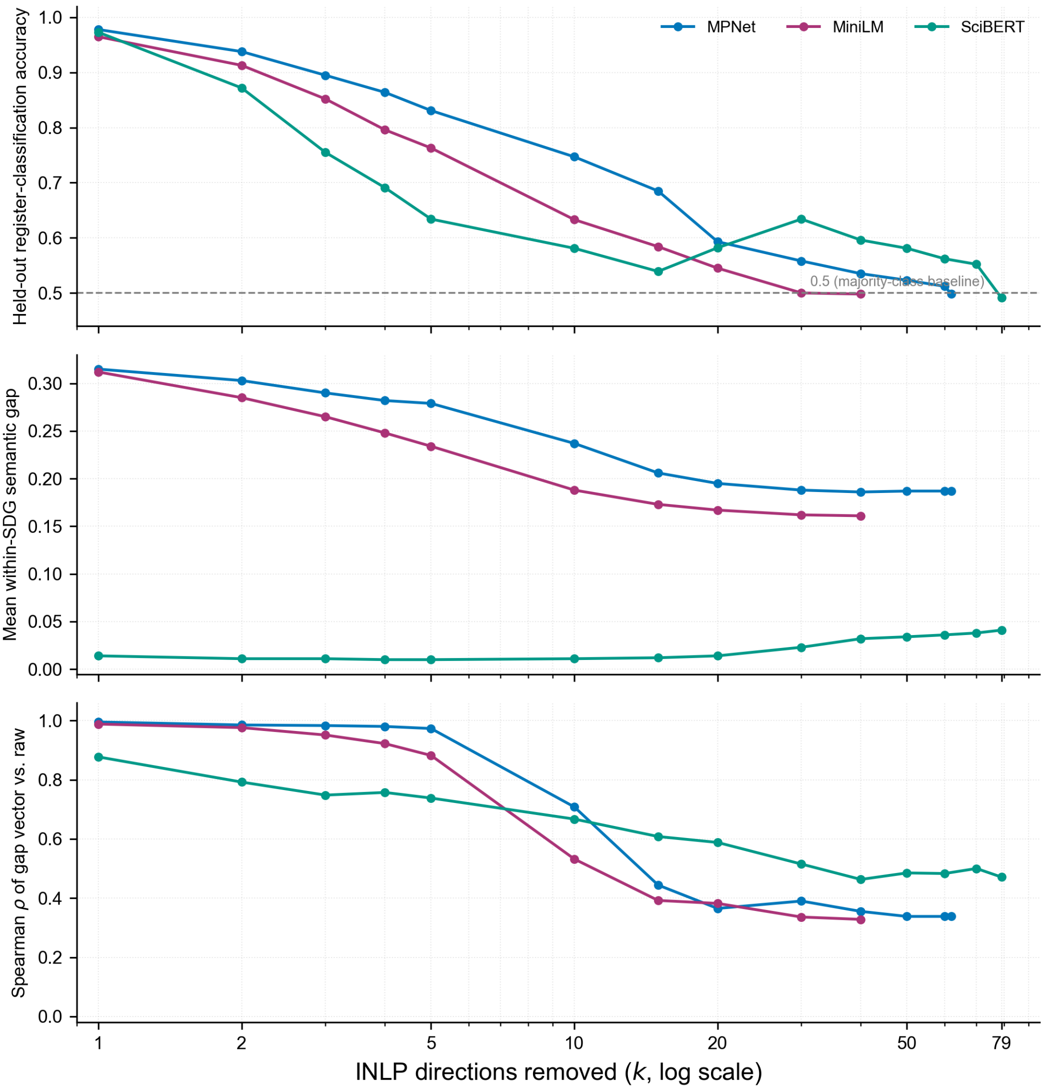

Figure 10. Stratified iterative register-removal diagnostic across the three encoder tracks.

Notes: Top: acc = held-out accuracy on a stratified 15% test split after projecting out the first k learned directions. Middle: gap = average within-SDG semantic gap after k removals. Bottom: p vs raw = Spearman rank correlation of the gap vector at iteration k versus the raw (iteration 0, un-projected) gap vector. Values are shown at the published iterations per encoder track; curves end where a track reached the stopping threshold.

G.2 Construct Validation

G.2.1 Purpose and Validation Question

Section 3.8 shows *what* INLP removes, a corpus-separable linear component, but not what that component is. This subsection evaluates the register interpretation empirically, using an independent surface-linguistic register score computed from the segment texts, following Biber-style register features (Biber, 1988).

The validation can show three things: whether the adjustment removes a large corpus-separable component; whether the removal is accompanied by evidence that SDG (topic) structure is being destroyed; and whether apparent residual register signals survive controlled samples. It cannot prove that the removed subspace is identical to the measured linguistic register features, and the register score used here is a small set of surface indicators. The validation is single-encoder and sample-limited.

G.2.2 Register Operationalisation and Sample Construction

What the test checks: The main analysis removes a writing-style (register) component with INLP and reads the remaining gap as topic. The appendix checks that reading with independent, surface-level style measurements. Two questions are asked: (1) does removal actually weaken the research–policy distinction? (2) does removal destroy topic? The short answer is yes to (1) and no to (2); whether the removed part is truly *register* is not proven (see the conclusion).

Sample construction: All results use the canonical MPNet track (12 segments per SDG per corpus, seed 42; 408 total) and the same frozen projection G. To avoid fake pooled correlations from long flagship reports (which sit close in embedding space), the main test uses one segment per source document (408 distinct parents); excluded flagship segments are replaced by shorter national/thematic monitoring reports from the same SDG (Table 12).

Table 12. Validation sample construction: the two designs.

Design	Segments	Research	Policy	Distinct docs	Multi-parent units	Mega-doc segments	Mega-SDGs
Original (per-SDG dedup)	408	204	204	390	25	25	15
One-per-parent (global dedup)	408	204	204	408	0	0	0

Notes: The one-per-parent sample is a rebuild, not a subset of the original sample: excluded flagship segments are replaced by shorter national and thematic monitoring reports, so the two samples differ in composition. Each sample holds 12 segments per SDG per corpus.

Register score: Each sampled segment is lower-cased and tokenised. Six surface features are counted and, except sentence length, expressed per 1,000 words:

- **Hedge rate:** occurrences of *may, might, could, suggests, appears, potentially, likely*.
- **Deontic-modality rate:** *must, shall, should, will*.
- **First-person rate:** first-person pronouns (*I, we, our, us, me, my* and variants).
- **Nominalisation rate:** words ending in *-tion, -ment, -ness*.
- **Passive-voice frequency:** a be-form (*am/is/are/was/were/be/been/being*) immediately followed by a past participle, allowing one intervening adverb; detected by part-of-speech tagging.
- **Mean sentence length:** words per sentence (sentence-split).

The register score is the first principal component (PC1) of the six standardised features. It is oriented so that it correlates positively with sentence length, but this is only a sign convention and the score does not rest on any single feature.

This manual score is the study's independent, human-readable register benchmark. It is not used in the main analysis, which relies on INLP's removed embedding subspace. It is the yardstick against which INLP removal is validated in G.2.3–G.2.5, where the manual score tracks INLP removal only weakly: its correlation with the amount of embedding removed is $\rho \approx 0.10$ on the original sample and not significant on the cleaner one-per-parent sample, and a classifier built from the six features alone separates research from policy at about chance. The manual score therefore captures only a small, unstable part of what INLP removes – which is why the register reading is offered as plausible but unproven.

Corpus feature contrasts: Table 13 shows all six features: policy uses markedly more deontic language and research hedges and passivises more, both stable across samples; first-person and nominalisation rates are about balanced, with the flagship segments the outliers on both. Mean sentence length is the one feature whose direction is not robust – its apparent policy lead is driven by the flagship documents and reverses once they are set aside, so it is treated as composition-sensitive rather than load-bearing.

G.2.3 Corpus Discrimination Before and After INLP

A 5-fold logistic-regression classifier separates research from policy segments almost perfectly on raw embeddings (accuracy 0.909 on the original sample; 0.944 on the one-per-parent sample; Table 15). After register removal the accuracy falls to 0.505 on the original sample and 0.603 on the one-per-parent sample – a large share of the corpus-linear signal is gone, though not all – and the canonical one-per-parent value stays above chance (0.603). Confidence intervals and p-values are in the table.

Table 13. Corpus register-feature contrasts (original sample).

Feature	Corpus mean		Direction	Load-bearing?	Mega-doc vs rest	
	Policy	Research			Mega-doc	Rest
Hedge rate (/1k words)	0.85	1.43	R > P	Yes	1.21	1.14
Deontic (must/should) rate (/1k words)	3.13	0.78	P > R	Yes	2.21	1.94
Passive-voice rate (/1k words)	8.45	10.30	R > P	Yes	3.35	9.77
Mean sentence length (words)	63.8	37.0	P > R	No	276.9	35.6
First-person rate (/1k words)	6.46	6.90	$\approx$	No	19.00	5.88
Nominalisation rate (/1k words)	39.00	36.37	$\approx$	No	20.02	38.84

Notes: Policy and Research are corpus-mean feature values on the original sample. Mega-doc = mean among the 25 flagship segments; Rest = mean among the remaining units. Rates are per 1,000 words except mean sentence length (words). Load-bearing? marks features used as stable corpus-style signals; sentence length is flagged composition-sensitive.

A per-SDG-deduplicated variant of the sample (dominated by the 25 flagship policy segments) yields an adjusted accuracy of 0.505, indistinguishable from chance. Excluding those segments raises it to 0.574 (difference +0.098, 95% CI [0.025, 0.169]), confirming that flagship clustering drives the original-sample result. The canonical one-per-parent estimate (0.603) is the primary result. Residual above-chance separation after removal is not a method failure, since INLP's stopping rule converges on its own training distribution while the validation classifier is re-estimated from scratch.

The register-only rows of Table 15 bound the interpretation. A classifier trained on the six surface features alone reaches only 0.456 on the original sample and 0.544 on the one-per-parent sample, barely above chance in the latter case. The linguistic markers therefore capture only a part of the corpus distinction. This is consistent with the register reading (the corpora do differ in register-like surface ways), but it also delimits it: the removed embedding component is far broader than these six surface features, so the validation cannot claim that the removed subspace is exhausted by, or identical to, the measured features.

G.2.4 Topic Preservation

If the adjustment removed topic, 17-way SDG selectivity, how well the SDG of a segment can be predicted from its embedding, would fall. It does not (Table 16): on the original per-SDG-dedup sample (draw 1), LR cross-validated accuracy is 0.691 on raw and 0.672 on adjusted embeddings, and kNN 0.554 and 0.578 (chance 0.059). The change is small and of mixed sign, so there is no evidence that topic structure is systematically removed; the analysis does not claim perfect preservation, and it was run on the original sample only, not re-estimated on the one-per-parent rebuild. Topic remains linearly decodable after register removal, as the identification argument anticipates.

G.2.5 Characterisation of the Removed Subspace

The tests above show that the removed component separates the corpora while preserving topic information. Two direction-space tests on the frozen SDG-balanced sample used to train G examine its structure.

Topic-direction orthogonality (P1). An independent 17-class SDG-topic logistic regression on the raw embeddings gives per-SDG weight vectors w_s. Alignment with $span(G)$ is $\|Gw_s\|/\|w_s\|$. The topic weights align with $span(G)$ at a mean of 0.248 (max 0.430), no more than a random 62-dimensional subspace (0.286, expected $\sqrt{62/768} \approx 0.284$); the corpus/register direction lies inside $span(G)$ at 0.979 (random 0.242). The removed subspace shows no preferential alignment with the independent SDG-topic directions, which argues against substantial SDG-topic leakage.

Held-out-SDG generalisation (P2). Leave-one-SDG-out, a corpus classifier trained on the other 16 SDGs still separates research from policy on the held-out SDG at 0.967 (full-data 0.978; held-out directions align with $span(G)$ at 0.981). These results argue against topic leakage. The direction generalises across unseen SDGs, which is consistent with a corpus-level component rather than an SDG-specific direction. A control topic direction (SDG 1 vs SDG 2) trains at 0.991 but classifies the other SDGs' research at only 0.781 (chance 0.5); a register direction trained on only SDG 1+2 still classifies the other 15 SDGs at 0.920, so the 0.967 is not a training-mass artefact (Table 14).

Lexical signature and interpretive limits. The removed subspace's word-level projection is weak and inconclusive. It does not provide a clean lexical fingerprint. In a full-vocabulary scan the maximum projection onto $span(G)$ is 0.527, versus 0.401 for a random 62-dimensional subspace. This modest gap indicates limited alignment with any specific vocabulary. The highest-scoring words combine a development–economy–governance cluster (*underdevelopment, economy, economically, infrastructure, governance, legislation, poverty, prosperity, socioeconomic, sustainable, liberalization, nationalize, decentralization, democratization*) together with *ai* and *convolutional* (apt for an AI-policy corpus), with a long tail of rare morphological variants (*developmentary, socionomics, governmentalize, reprivatization, deindustrialization*) and obscure lexicon items (*microtasimeter, sciuroid, astrolabe, decipium*) that score highly for incidental embedding reasons; SDG leaf-topics (climate, water, health, biodiversity) are absent from the top words. The full ranked list (top 200 words and the random-subspace null) is in the committed scan file. This word-level evidence does not establish a policy-domain vocabulary. P1 and P2 support the narrower interpretation that the removed subspace captures corpus-associated variation that is partly register-like. Embedding spaces encode word content on their dominant linear directions (Conneau et al., 2018; Alain and Bengio, 2018; Ethayarajh, 2019), and register is entangled with topic and domain (Grieve et al., 2025; John et al., 2019); the corpus-separating direction INLP learns therefore reflects prevalent cross-cutting vocabulary (development, economic, governance) rather than genre. The projection also differs from the Biber surface-register features targeted by the manual score in G.2.2: that score tracks G only weakly ($\rho \approx 0.10$).

Implementation trust. The INLP loop that produced G was verified bit-for-bit against Ravfogel et al.'s reference implementation: principal-angle cosines between the two removed subspaces are 1.0 and a fresh classifier collapses to chance after either projection (promoted parity check at `1_code/_inlp_parity/`; synthesis in `5_notes/register_validity/V` §6). The P1/P2 results are trustworthy as INLP results.

Scope. These tests are MPNet-canonical only; MiniLM and SciBERT were not re-verified under P1/P2.

Table 14. Direction-space characterisation of the removed subspace G (MPNet canon).

Test	Quantity	Value	Comparison
P1	Topic-weight align. w/ $span(G)$ (mean)	0.248	random 0.286
P1	Topic-weight align. w/ $span(G)$ (max)	0.430	—
P1	Corpus/register align. w/ $span(G)$	0.979	random 0.242
P2	Held-out-SDG corpus acc. (mean)	0.967	full-data 0.978
P2	Topic-control dir. on other SDGs	0.781	chance 0.5
P2	Register dir. (2 SDG) on other 15	0.920	train 0.976
P0b	Max vocab projection on $span(G)$	0.527	random 0.401

Notes: P1 aligns independent SDG-topic and corpus logistic-regression weight vectors with $span(G)$ (orthonormal rows, so alignment $= \|Gw\|/\|w\|$); a random 62-dimensional subspace is the null. P2 is leave-one-SDG-out corpus-classification accuracy; the topic control is an SDG 1-vs-2 direction's accuracy on other SDGs' research (chance 0.5), and the balanced control is a register direction trained on only SDG 1+2 tested on the other 15 SDGs. P0b calibration is the maximum projection norm of the full English vocabulary onto $span(G)$ versus a random subspace. All figures reproduce from frozen embeddings and G under seed 42.

G.2.6 Residual Register Diagnostics and Robustness Investigation

The register score used here is the independent surface-linguistic score defined in G.2.2. Does any register survive removal? Two faint traces appear on the messy sample but both vanish or reverse on the cleaner one-per-parent sample, so they reflect flagship-document clustering, not real residual register. Specifically, on the messy sample the manual score correlates weakly with the amount removed (+0.102, within-research +0.212, within-policy +0.191), and correlates more strongly with SDG-centroid distance in adjusted than raw space (+0.126 $\rightarrow$ +0.247). On the clean sample both correlations disappear or go negative (+0.092; centroid -0.212 $\rightarrow$ -0.197). A within-SDG trace (-0.197) and the draw-stability check (seeds 43–45, sign-flipping, only 0/17 SDGs significant) confirm it is draw-level noise, not structure.

G.2.7 Validation Conclusion

The validation supports the following claims.

- The adjustment removes a corpus-level component: corpus-discrimination accuracy falls from near-perfect on raw embeddings to moderately above chance after removal, on both samples (Table 15).
- The removed component is not SDG topic loss: 17-way SDG selectivity is essentially unchanged (Table 16), and SDG-topic weight vectors are orthogonal to $span(G)$ while the corpus/register direction lies inside it (P1; Table 14).
- The register direction generalises across held-out SDGs at 0.967 (P2), falsifying the topic-leakage threat; the removed subspace is topic-independent at the SDG level.
- Apparent residual register signals were sampling/document-clustering effects, not stable residual structure: the removed-magnitude and centroid-distance associations vanished or reversed on the controlled sample, and the one surviving within-SDG trace was draw-unstable.

The structural claim is supported in qualified form: the removed subspace carries corpus-associated information, shows no preferential alignment with SDG-topic directions, and generalises across held-out SDGs. These results support interpreting it as partly register-like, while they do not establish a pure register axis. A full-vocabulary scan (P0b) places the top projection only modestly above a random subspace (0.527 vs 0.401), and the highest-scoring words combine socioeconomic and governance terms with rare morphological items. The word list does not establish a policy-domain vocabulary.

G.2.8 Limitations

The validation is bounded in ways that should be read alongside the main text. First, the register score is one choice among several: an alternative a-priori “institutional” combination gave null or reversed results, and the score’s sign is a convention, not a substantive prior. Second, each sample contains only 408 segments (12 per SDG per corpus), and the per-SDG tests within the draw-stability stage have $n = 12$ policy units each, so the 0-of-17 significant per-SDG result is not evidence of absence at the per-SDG level. Third, the draw-stability check was applied to the within-SDG trace only; the corpus-level accuracies in Table 15 were not re-estimated on fresh draws. Fourth, the one-per-parent sample is a rebuild, not a subset of the original sample, and the excluded flagship documents are replaced by shorter national and thematic monitoring reports, so the two-sample comparison is partly compositional. Fifth, the validation is MPNet-canonical only; P1/P2 likewise. Sixth, the residual corpus-linear signal is characterised at the direction level (P1/P2). The word-level lexical signal (P0b) is weak and noisy, so it does not establish a policy-domain vocabulary. INLP removes only linear information, so non-linear register structure remains (Ravfogel et al., 2020; Bolukbasi et al., 2016; Elazar, Jacovi, and Goldberg, 2021; Gonen and Goldberg, 2019). Register is also entangled with topic (Grieve et al., 2025), so the removed direction is best treated as corpus-associated and register-like rather than as a pure register axis. Open only whether a finer sub-lexical decomposition would recover more structure.

Table 15. Corpus discrimination accuracy (research vs. policy, 5-fold logistic regression) before and after INLP register removal.

Sample	Input	n	Fold-mean acc.	Pooled acc. (k/n)	Wilson 95% CI	Binom. p
Original (per-SDG dedup)	register-only	408	0.456	0.456 (186/408)	[0.408, 0.504]	0.967
Original (per-SDG dedup)	raw embeddings	408	0.909	0.909 (371/408)	[0.878, 0.933]	< 0.001
Original (per-SDG dedup)	adjusted (INLP)	408	0.505	0.505 (206/408)	[0.457, 0.553]	0.441
One-per-parent	register-only	408	0.544	0.544 (222/408)	[0.496, 0.592]	0.042
One-per-parent	raw embeddings	408	0.944	0.944 (385/408)	[0.917, 0.962]	< 0.001
One-per-parent	adjusted (INLP)	408	0.603	0.603 (246/408)	[0.555, 0.649]	< 0.001
Original (per-SDG dedup)	adjusted, mega-policy excluded	383	0.574	0.574 (220/383)	[0.524, 0.623]	0.002

Notes: Pooled accuracy = proportion of correct predictions across the five test folds. Wilson 95% CI and one-sided binomial test against 0.5 on the pooled proportion. In the one-per-parent sample each segment comes from a distinct source document, so no test segment shares a source document with its training folds; in the original per-SDG-dedup sample, segments of the same flagship document can straddle folds, so the original-sample rows carry a mild grouped-similarity caveat. Mega-policy excluded = original sample minus the 25 policy segments contributed by the 7 multi-segment documents.

Table 16. 17-way SDG (topic) selectivity before and after register removal (original per-SDG-dedup sample, draw 1).

Input	LR CV acc.	kNN(5) CV acc.	Chance
Raw	0.691	0.554	0.059
Adjusted (INLP)	0.672	0.578	0.059

Notes: Five-fold stratified cross-validated accuracy of logistic regression ($C = 10$) and k-nearest-neighbour classification ($k = 5$) (Cover and Hart, 1967) in predicting the SDG of each segment from raw and adjusted embeddings; chance = 1/17. If the adjustment removed topic, these accuracies would fall.

H SUPPLEMENTARY CROSS-METHOD DATA

H.1 Cross-Method Coverage and Semantic Gap Values

Figure 11 reports the raw coverage gap and semantic gap values for each SDG across seven encoder–classifier combinations, with concept-retrieval LR and MLP columns for comparison against the keyword-retrieved research corpus baseline (Appendix A). While Tables 17 and 22 rank these values to compare cross-method stability, the figure annotates the gap magnitudes per cell, allowing the rank separation to be interpreted in absolute terms. Coverage gap is document-weighted (Section 3.6). Semantic gap policy centroids use a 50-segment-per-document cap. The zero-shot nearest-centroid method is reported for the canonical MPNet encoder in the rank tables, the value figure, and Appendix H.5. The pooled regression of Appendix I additionally includes it as an assignment-method configuration under each encoder.

H.2 Rank Robustness: Cross-Sensitivity and Encoder Sensitivity

Tables 17 (adjusted) and 18 (raw) report each SDG's semantic-gap rank under the canonical MPNet–LR configuration and its alternatives: segment cap (20, none) and concept-based retrieval (LR, MLP) in both panels, with policy source family (curated AI/SDG, SDGi, UNGDC, full) added in the raw panel. Tables 19, 20, and 21 report the within-SDG gap rankings under MPNet, MiniLM, and SciBERT for each classifier. The adjusted (register-removed) ranking is reported separately from the raw (naive) ranking, with the tested dimension nested inside each table.

H.3 Encoder Sensitivity

The semantic-gap ranking is moderately preserved across the three encoders on the adjusted (register-removed) gap (MPNet–LR vs. SciBERT–LR Spearman $\rho = 0.537$; MPNet–LR vs. MiniLM–LR $\rho \approx 0.632$), and the coverage-gap ranking is highly stable (MPNet–LR vs. SciBERT–LR $\rho = 0.934$). MiniLM and SciBERT are scored on a deterministic 100,000-paper subset (seed 42) of the canonical segmented corpus, so the architecture comparison isolates encoder choice from corpus scale. Under the raw gap, SDG 3 leads for MPNet–LR and SciBERT–LR while SDG 13 leads for MiniLM–LR; SDG 17 trails in all three (rank 16 or 17). The scientific-domain SciBERT compresses the magnitudes but does not change which goals diverge most. Classifier sensitivity within MPNet on the adjusted gap: the three classifiers, LR, zero-shot nearest-centroid, and the MLP robustness check, show varied agreement (MPNet LR vs. MLP $\rho = 0.850$, LR vs. zero-shot $\rho = 0.596$). Under the raw gap, SDG 17 is the most classifier-sensitive goal, sitting at rank 17 under LR but rank 8 under zero-shot. The LR baseline is the canonical reference, with MLP and zero-shot as uncertainty bounds.

H.4 Per-Track Reference Value Tables

The per-SDG reference table from Section 4.4 is reproduced below for the MiniLM and SciBERT encoder tracks (each scored on the deterministic 100,000-paper subset, isolating encoder choice from corpus scale). The table reports a semantic-gap family (raw, adjusted, register component) and a coverage-predictor family (coverage gap, signed dominance, research share, policy share). Signed dominance = research share $-$ policy share (pp); coverage gap = $|$research $-$ policy$|$ (pp).

H.5 Assignment-Method Comparison: Supervised vs. Nearest-Centroid

The zero-shot nearest-centroid method is used in this study for one purpose only: a single comparison between the canonical supervised classifier (LR, with the MLP as robustness) and an assignment rule with no trained decision boundary. Table 25 reports how often the two classifiers agree, per SDG and overall, for the research corpus (segment level) and the policy corpus (segment level), together with the LR-versus-zero-shot semantic-gap rank delta per SDG.

The comparison confirms that the two classifiers agree on the majority of assignments: the overall LR-versus-zero-shot agreement rate is 67.3% over the 3,105,144 research segments and 81.5% over policy segments (Table 25). Agreement is not uniform across SDGs, and the disagreements are concentrated where the reference centroids are least separated, in line with Appendix B.2. SDG 17, whose reference centroid is the flattest and most isolated in that analysis, is also the most classifier-sensitive goal here: it shows the largest rank delta between the two semantic-gap rankings ($|\Delta| = 9$) and one of the lowest per-SDG classifier agreement rates on the research side (26.1%, second only to SDG 10 at 23.6%). SDG 8, whose supervised rank is also heavily classifier-dependent (rank 6 under LR, rank 14 under zero-shot), shows one of the largest rank deltas ($|\Delta| = 8$, tied with SDG 9). The zero-shot rule, having no supervised decision boundary, anchors partnership and inequality discourse less sharply than the LR classifier. The MLP robustness check agrees with the zero-shot rule at 76.3% of policy segments, close to the LR-versus-zero-shot rate, and at 59.5% of research segments overall.

H.6 Raw-Value Correlation: Metric-Sensitivity Companion

Table 26 repeats the H1a–H1d grid of Table 4 with one change: cells report Pearson r on the raw coverage and gap magnitudes instead of Spearman ρ on their ranks. The grid, predictors, and columns are identical. The four H1 hypotheses are co-equal and each probes a distinct coverage-structure prediction. Under both metrics the adjusted-gap association is stable across the supervised tracks for H1a and H1d (the hypotheses with a detectable signal: positive in 8/9 and 9/9 configs respectively under Pearson) and remains null for H1b and H1c (no consistent signal under either metric – itself informative: coverage divergence tracks the gap and policy share, not dominance or raw research share). The zero-shot config, already scoped as an uncertainty bound, is stable on the adjusted gap. SciBERT is designated a lower-trust robustness encoder (its scientific pretraining compresses gap magnitudes, 0.043 against 0.340, Section H.3).

Possible causes of the SciBERT divergence. Possible causes range from small-n estimation noise to SciBERT's compressed gap magnitudes and lower assignment fidelity. None is confirmed, and the divergence does not threaten the conclusion, which rests on the supervised tracks where the H1a adjusted Pearson reaches $+0.694$ (Table 26) and is stable across metrics. The raw-value specification therefore confirms, rather than threatens, the rank-based result.

H.7 Detailed Policy Profiles by Source Family

The full policy corpus combines a curated AI/SDG sub-corpus, SDGi VNR/VLR reports, and UN General Debate speeches. These are all policy-facing texts, but they are not interchangeable genres. The family split reported in Table 27 therefore asks a narrow question: whether the full-corpus policy profile is broadly stable across source families or whether it is strongly shaped by corpus composition.

The answer is mixed. The international-institutional profile is source-family sensitive. The curated AI/SDG family is much more innovation-oriented, with SDG 9 leading its document-weighted profile, while the UNGDC family is more strongly concentrated on climate, partnerships, and governance, consistent with the UNGD topic distributions reported by Jankin, Baturo, and Dasandi (2024), and the SDGi family is comparatively even, with SDG 11-style institutional review language at its top. At the same time, the broader full-corpus pattern is not driven by one family alone: both the UNGDC speeches and the full corpus are dominated by the climate–partnerships–governance block, and the within-SDG semantic-gap ordering is broadly preserved across families ($\rho = 0.73$–0.97).

The implication is interpretive rather than corrective. The full policy corpus should be read as international institutional SDG discourse, not as a universal profile of all AI sustainability policy. The scope of what the policy-side distribution represents is accordingly narrower, but the main diagnostic purpose remains valid.

Re-running the interaction within each policy family (reliable SDGs only; Table 28) confirms the null is not a scope-mismatch artefact: the curated AI/SDG family reproduces the full-corpus pattern, while SDGi ($\rho = 0.596$, $p = 0.014$) and UNGDC ($\rho = 0.425$, $p = 0.115$) show positive research-share associations. The register-adjusted coverage-gap association is positive for SDGi ($\rho = 0.498$, $p = 0.044$) and full corpus ($\rho = 0.544$, $p = 0.025$) but null under symmetric scope (curated $\rho = -0.061$, $p = 0.807$; UNGDC $\rho = 0.357$, $p = 0.191$), bounding its reach. Two caveats: the curated family's narrower share range attenuates rank correlations, and the register-removal caveat of Appendix G.2 applies.

Table 17. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG		Segment cap		Retrieval	
	Canon	20	None	LR	MLP
1	8	8	8	4	6
2	16	16	17	17	17
3	2	1	2	3	5
4	11	11	11	14	14
5	13	13	13	6	10
6	15	15	15	15	16
7	10	10	9	13	13
8	9	9	7	5	1
9	4	5	3	7	7
10	7	7	10	9	3
11	6	6	5	2	2
12	12	12	12	10	11
13	5	4	4	11	12
14	14	14	14	12	9
15	17	17	16	16	15
16	3	3	6	8	8
17	1	2	1	1	4
Rank Corr (ρ)	1.000	0.995	0.966	0.755	0.681

Adjusted (register-removed, canonical).

Notes: Each cell reports the within-SDG semantic-gap rank ($1 =$ largest gap, $17 =$ smallest gap). The Canon column is the canonical MPNet—LR ranking. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the Canon column.

Table 18. Cross-sensitivity robustness of within-SDG semantic-gap rankings across policy source, segment cap, and retrieval strategy.

SDG		Policy source				Segment cap		Retrieval	
	Canon	Curated	SDGi	UNGDC	Full	20	None	LR	MLP
1	5	2	5	5	5	5	5	1	1
2	10	10	11	11	10	10	9	11	10
3	1	8	1	1	1	1	2	3	7
4	15	15	13	16	15	15	14	13	17
5	13	13	14	13	13	13	13	6	5
6	12	9	12	12	12	12	12	15	15
7	4	1	6	2	4	4	4	8	11
8	6	5	7	6	6	6	6	4	2
9	9	14	2	10	9	8	8	12	13
10	14	11	15	15	14	14	15	10	8
11	3	4	3	4	3	3	3	2	3
12	11	6	10	8	11	11	11	14	12
13	2	3	4	3	2	2	1	5	4
14	7	7	9	9	7	7	7	9	6
15	16	12	16	14	16	16	16	16	16
16	8	16	8	7	8	9	10	7	9
17	17	17	17	17	17	17	17	17	14
Rank Corr (ρ)	1.000	0.733	0.914	0.966	1.000	0.998	0.988	0.814	0.679

Raw (naive baseline, un-adjusted).

Notes: Each cell reports the within-SDG semantic-gap rank ($1 =$ largest gap, $17 =$ smallest gap). The Canon column is the canonical MPNet—LR ranking. Policy-source columns compare the keyword-retrieved research profile against each policy-source family. Segment-cap columns compare cap 20 and no cap. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the Canon column.

Table 19. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	**8**	**11**	**1**	**8**	**7**	**15**	**1**
2	**16**	**16**	**14**	**17**	**17**	**17**	**16**
3	**2**	**1**	**3**	**2**	**1**	**5**	**12**
4	*11*	*14*	*13*	*7*	*8*	*9*	*11*
5	*13*	*10*	*11*	*1*	*3*	*2*	*6*
6	**15**	**15**	**10**	**16**	**15**	**14**	**17**
7	10	12	8	12	12	7	10
8	9	4	12	11	11	13	14
9	*4*	*9*	*16*	*10*	*9*	*11*	*15*
10	7	5	4	4	4	6	2
11	*6*	*6*	*9*	*13*	*13*	*4*	*5*
12	12	7	5	9	10	16	8
13	**5**	**8**	**6**	**6**	**5**	**3**	**4**
14	**14**	**13**	**15**	**15**	**14**	**10**	**7**
15	17	17	17	14	16	12	13
16	**3**	**3**	**7**	**3**	**2**	**1**	**3**
17	*1*	*2*	*2*	*5*	*6*	*8*	*9*
Rank Corr (ρ)	1.000	0.850	0.596	0.632	0.713	0.537	0.363

Adjusted (register-removed, canonical).

Notes: Each cell reports the within-SDG semantic-gap rank ($1 =$ largest gap, $17 =$ smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix H.5). Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical adjusted MPNet-LR column.

Table 20. Domain-encoder sensitivity of within-SDG semantic-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	*5*	*8*	*7*	*17*	*17*	*16*	*1*
2	*10*	*11*	*10*	*14*	*14*	*14*	*14*
3	**1**	**4**	**4**	**2**	**2**	**1**	**3**
4	**15**	**15**	**15**	**15**	**15**	**10**	**12**
5	*13*	*9*	*16*	*4*	*8*	*11*	*15*
6	**12**	**14**	**5**	**13**	**16**	**6**	**8**
7	**4**	**7**	**1**	**5**	**4**	**4**	**4**
8	*6*	*3*	*14*	*10*	*12*	*15*	*11*
9	**9**	**12**	**17**	**8**	**6**	**7**	**13**
10	*14*	*13*	*12*	*3*	*3*	*5*	*7*
11	3	2	2	6	7	3	2
12	*11*	*5*	*9*	*7*	*5*	*13*	*5*
13	**2**	**1**	**6**	**1**	**1**	**2**	**6**
14	*7*	*6*	*3*	*11*	*10*	*8*	*9*
15	*16*	*17*	*11*	*9*	*13*	*9*	*10*
16	*8*	*10*	*13*	*12*	*11*	*12*	*16*
17	**17**	**16**	**8**	**16**	**9**	**17**	**17**
Rank Corr (ρ)	1.000	0.863	0.534	0.400	0.373	0.463	0.618

Raw (naive baseline, un-adjusted).

Notes: Each cell reports the within-SDG semantic-gap rank ($1 =$ largest gap, $17 =$ smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only (scoped to a single supervised-vs-nearest-centroid comparison, Appendix H.5). Rank Corr (ρ) is the Spearman correlation of each column's SDG gap ranks against the canonical raw MPNet-LR column.

Table 21. Domain-encoder sensitivity of within-SDG coverage-gap rankings across embedding architectures of differing domain specialisation but matched dimensionality.

SDG	Encoder (embedding architecture)						
	MPNet (768-d)			MiniLM (384-d)		SciBERT (768-d)	
	LR	MLP	ZS	LR	MLP	LR	MLP
1	**7**	**6**	**7**	**7**	**7**	**7**	**6**
2	*15*	*13*	*14*	*11*	*11*	*11*	*9*
3	**1**	**1**	**2**	**1**	**1**	**1**	**1**
4	**4**	**4**	**3**	**5**	**4**	**3**	**3**
5	*8*	*10*	*8*	*12*	*9*	*8*	*8*
6	**9**	**8**	**12**	**8**	**10**	**12**	**10**
7	**16**	**17**	**6**	**16**	**17**	**17**	**17**
8	11	15	9	9	8	9	11
9	**2**	**2**	**4**	**2**	**2**	**2**	**4**
10	**14**	**9**	**11**	**15**	**14**	**15**	**14**
11	*17*	*16*	*15*	*13*	*13*	*14*	*13*
12	12	14	13	14	15	13	15
13	**5**	**5**	**5**	**6**	**6**	**6**	**5**
14	**10**	**12**	**16**	**10**	**12**	**10**	**12**
15	*13*	*11*	*17*	*17*	*16*	*16*	*16*
16	6	7	10	3	3	5	7
17	**3**	**3**	**1**	**4**	**5**	**4**	**2**
Rank Corr (ρ)	1.000	0.919	0.748	0.895	0.902	0.934	0.897

Notes: Each cell reports the within-SDG coverage-gap rank ($1 =$ largest gap, $17 =$ smallest gap) under that encoder and assignment method. MPNet (768-d) is the canonical general-purpose encoder; MiniLM (384-d) is a smaller general-purpose encoder; SciBERT (768-d) is a scientific-domain encoder. MPNet and SciBERT share dimensionality (768-d), isolating architecture/domain from the dimensionality drop that confounds the MPNet–MiniLM pair (Section H.3). LR = canonical supervised logistic regression; MLP = 4-layer/384-hidden network; Zero-shot = nearest-centroid on SDG reference centroids, reported for the canonical MPNet encoder only. Coverage gap is document-weighted (Assumption A19) for all methods. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical MPNet-LR column.

Table 22. Cross-sensitivity robustness of within-SDG coverage-gap rankings across policy source and retrieval strategy.

SDG		Policy source			Retrieval	
	Canon	Curated	SDGi	UNGDC	LR	MLP
1	7	17	4	11	6	6
2	15	8	12	12	12	13
3	1	3	1	2	2	2
4	4	4	3	6	7	14
5	8	11	7	17	8	10
6	9	15	9	16	9	8
7	16	7	17	8	17	17
8	11	9	8	9	15	15
9	2	1	2	4	1	1
10	14	16	13	15	14	9
11	17	5	15	7	10	7
12	12	14	10	10	16	16
13	5	6	16	5	5	5
14	10	13	14	13	11	11
15	13	10	11	14	13	12
16	6	12	6	1	4	4
17	3	2	5	3	3	3
Rank Corr (ρ)	1.000	0.397	0.775	0.544	0.868	0.662

Notes: Each cell reports the within-SDG coverage-gap rank ($1 =$ largest gap, $17 =$ smallest gap). The Canon column is the canonical MPNet—LR ranking. Coverage gap = $|$research proportion $-$ policy proportion$|$ per SDG, using document-weighted policy proportions (Assumption A19). Policy-source columns compare the keyword-retrieved research profile against each policy-source family. The Retrieval column replaces keyword retrieval with concept-based (AI/ML field-of-study) retrieval. The Segment-cap axis is omitted: coverage gap is segment-cap-independent. Rank Corr (ρ) is the Spearman correlation of each column's SDG coverage-gap ranks against the canonical column.

Table 23. MiniLM track — per-SDG reference values (semantic-gap and coverage-predictor families).

SDG	Description	Semantic gaps			Coverage predictors			
		Raw	Adj.	Reg.	Cov.	SDom	Res %	Pol %
1	No Poverty	0.222	0.176	+0.046	4.1	-4.1	0.3	4.4
2	Zero Hunger	0.297	0.077	+0.220	2.1	-2.1	2.3	4.4
3	Good Health	0.495	0.238	+0.257	22.0	+22.0	26.9	4.9
4	Education	0.291	0.178	+0.113	8.9	+8.9	13.8	4.9
5	Gender Equality	0.470	0.272	+0.198	1.7	-1.7	3.3	5.1
6	Clean Water	0.298	0.101	+0.198	2.3	-2.3	1.4	3.7
7	Clean Energy	0.430	0.127	+0.302	1.0	+1.0	5.3	4.3
8	Decent Work	0.336	0.129	+0.207	2.3	-2.3	2.0	4.2
9	Innovation	0.350	0.142	+0.208	12.5	+12.5	16.9	4.4
10	Reduced Ineq.	0.485	0.207	+0.278	1.1	-1.1	1.8	2.9
11	Sust. Cities	0.397	0.119	+0.278	1.6	-1.6	3.7	5.3
12	Consumption	0.391	0.147	+0.245	1.6	-1.6	1.9	3.5
13	Climate	0.541	0.187	+0.354	5.1	-5.1	5.6	10.7
14	Life Below Water	0.330	0.111	+0.218	2.2	-2.2	1.4	3.6
15	Life on Land	0.345	0.112	+0.233	0.1	-0.1	4.1	4.2
16	Peace & Justice	0.322	0.220	+0.101	11.2	-11.2	8.7	19.9
17	Partnerships	0.241	0.190	+0.052	8.9	-8.9	0.6	9.5
Mean	—	0.367	0.161	+0.206	5.2	+0.0	5.9	5.9

Notes: The semantic-gap and coverage-predictor structure is retained. Signed dom. = research % − policy % (pp); Reg. Comp. = raw − adjusted.

Table 24. SciBERT track — per-SDG reference values (semantic-gap and coverage-predictor families).

SDG	Description	Semantic gaps			Coverage predictors			
		Raw	Adj.	Reg.	Cov.	SDom	Res %	Pol %
1	No Poverty	0.032	0.025	+0.007	5.1	-5.1	0.1	5.3
2	Zero Hunger	0.037	0.020	+0.017	2.6	-2.6	1.3	3.9
3	Good Health	0.056	0.044	+0.012	22.3	+22.3	27.0	4.8
4	Education	0.044	0.039	+0.004	12.8	+12.8	17.7	4.9
5	Gender Equality	0.041	0.063	-0.022	3.9	-3.9	1.5	5.4
6	Clean Water	0.045	0.028	+0.017	2.5	-2.5	1.1	3.6
7	Clean Energy	0.052	0.042	+0.010	0.6	-0.6	3.5	4.2
8	Decent Work	0.033	0.030	+0.004	3.7	-3.7	0.8	4.5
9	Innovation	0.045	0.035	+0.010	17.6	+17.6	21.4	3.8
10	Reduced Ineq.	0.047	0.043	+0.004	1.6	-1.6	0.8	2.4
11	Sust. Cities	0.052	0.054	-0.001	1.8	-1.8	4.1	5.9
12	Consumption	0.039	0.024	+0.014	2.5	-2.5	1.1	3.5
13	Climate	0.053	0.056	-0.003	5.9	-5.9	4.0	9.9
14	Life Below Water	0.044	0.039	+0.005	2.8	-2.8	0.9	3.7
15	Life on Land	0.044	0.032	+0.012	1.3	-1.3	2.8	4.1
16	Peace & Justice	0.040	0.078	-0.037	9.1	-9.1	11.6	20.7
17	Partnerships	0.023	0.041	-0.019	9.1	-9.1	0.2	9.3
Mean	—	0.043	0.041	+0.002	6.2	+0.0	5.9	5.9

Notes: The semantic-gap and coverage-predictor structure is retained. Signed dom. = research % − policy % (pp); Reg. Comp. = raw − adjusted.

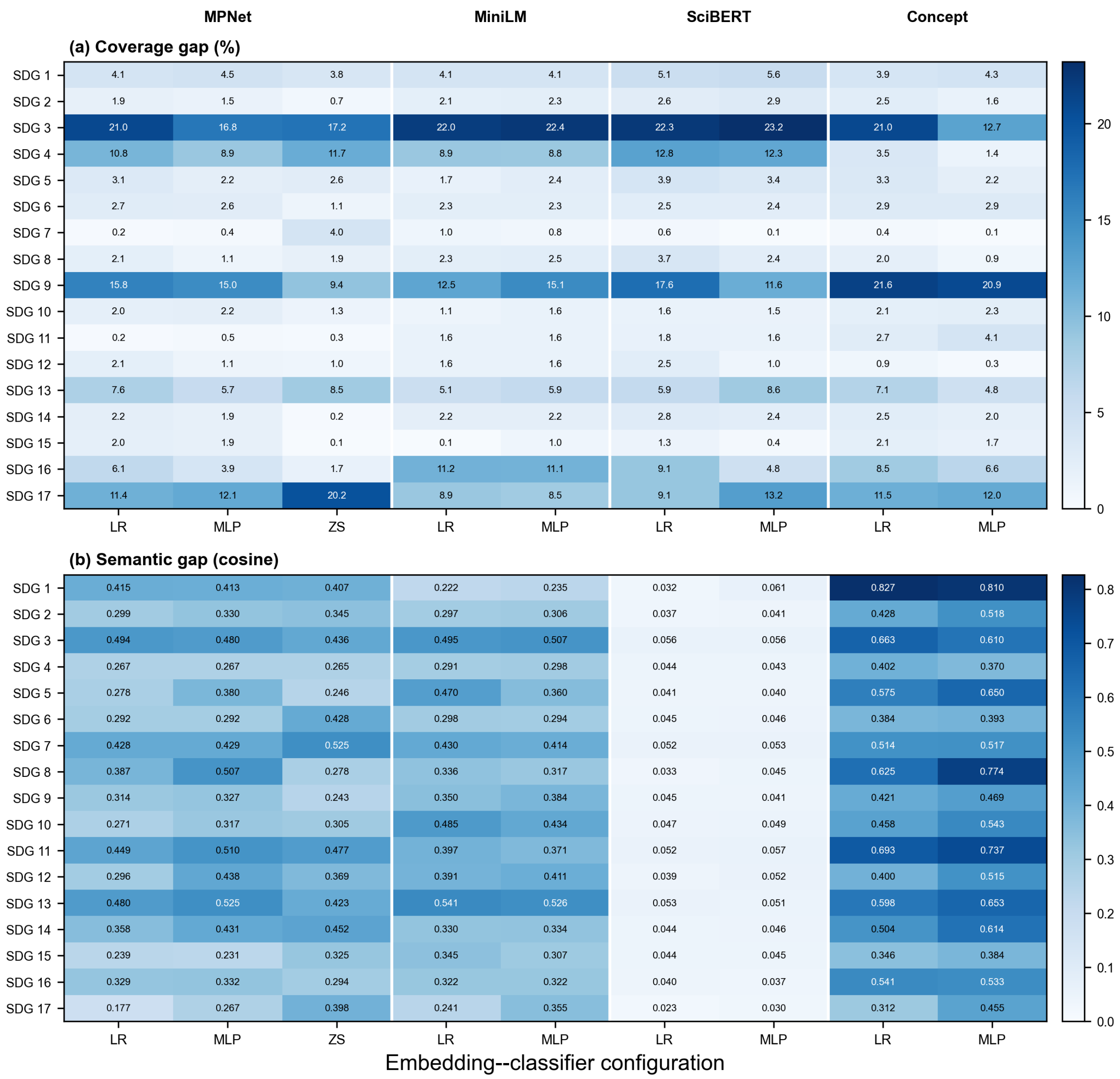

Figure 11. Coverage gap values and semantic gap values across embedding architectures, classifier variants, and retrieval strategies.

Notes: Top: coverage gap = |research proportion − policy proportion| per SDG in percentage points, using document-weighted policy proportions (Section 3.6) for all methods. Bottom: semantic gap = $1 - \cos(\mathbf{r}_{\text{SDG}}, \mathbf{p}_{\text{SDG}})$ where $\mathbf{r}_{\text{SDG}}$ and $\mathbf{p}_{\text{SDG}}$ are the research and policy centroids; policy centroids use a 50-segment-per-document cap, values in cosine units. This is a *values* figure reporting the raw (naive) semantic gap; the adjusted (register-removed) ranking is shown in the rank-robustness tables (Section H.2).

Table 25. Classifier agreement between the supervised LR classifier and the zero-shot nearest-centroid rule, by SDG and overall, under the canonical MPNet encoder and keyword retrieval.

	Research (segments)			Policy (segments)		Semantic-gap rank		
SDG	n_{LR}	Agree %	Res MLP %	n_{LR}	Agree %	LR	ZS	$\lvert\Delta\rvert$
1 (No Poverty)	13,459	30.9	14.9	1,700	70.8	5	7	2
2 (Zero Hunger)	78,591	77.6	59.8	1,723	94.1	10	10	0
3 (Good Health)	859,355	75.0	79.8	2,934	87.7	1	4	3
4 (Education)	470,002	82.4	82.6	2,055	91.3	15	15	0
5 (Gender Equality)	64,940	56.3	37.0	2,026	88.5	13	16	3
6 (Clean Water)	35,137	85.3	75.2	1,206	94.6	12	5	7
7 (Clean Energy)	119,467	88.3	81.9	1,808	84.1	4	1	3
8 (Decent Work)	65,500	44.4	20.7	2,016	71.3	6	14	8
9 (Innovation)	577,833	50.8	45.1	4,066	81.0	9	17	8
10 (Reduced Ineq.)	30,323	23.6	12.8	1,123	56.9	14	12	2
11 (Sust. Cities)	146,508	77.2	64.0	1,658	88.3	3	2	1
12 (Consumption)	46,199	60.9	38.5	1,187	91.8	11	9	2
13 (Climate)	121,557	40.5	28.4	3,399	84.5	2	6	4
14 (Life Below Water)	51,458	89.6	69.7	1,634	92.3	7	3	4
15 (Life on Land)	78,102	83.5	70.3	1,828	84.1	16	11	5
16 (Peace & Justice)	337,317	55.6	46.1	5,725	65.0	8	13	5
17 (Partnerships)	9,396	26.1	11.7	4,509	84.1	17	8	9
Overall	3,105,144	67.3	59.5	40,597	81.5			

Notes: Agree % is the share of LR-assigned texts that the zero-shot rule also assigns to the same SDG. Semantic-gap ranks are $1 =$ largest gap on the raw (naive, unadjusted) gap; $|\Delta|$ is the absolute rank difference between the two classifiers. Classifier agreement is computed over all texts (research: each segment; policy: each segment), without the 50-segment-per-document cap, which applies only to centroid building. The zero-shot rule assigns each text to the SDG with the highest cosine similarity to the train-split reference centroids $(\mathbf{s}_k)$; the LR rule uses the trained classifier's argmax. Research MLP-vs-ZS agreement is computed from the persisted per-shard research scores and shown as the overall Res MLP % (59.5%). The policy-side MLP rate appears in the text below.

Table 26. Pearson r on the raw coverage and semantic-gap magnitudes (not ranked).

	Semantic Gap	Adj. gap	Reg. Comp.
H1a. Coverage gap $\leftrightarrow$ Semantic gap			
MPNet LR	+0.102	+0.694**	-0.382
MPNet MLP	-0.069	+0.441†	-0.404
MPNet ZS	-0.171	+0.230	-0.333
MiniLM LR	+0.068	+0.474†	-0.226
MiniLM MLP	+0.333	+0.611**	-0.093
SciBERT LR	+0.129	+0.152	-0.080
SciBERT MLP	-0.108	-0.189	+0.138
Concept LR	+0.030	+0.445†	-0.357
Concept MLP	-0.092	+0.196	-0.310
H1b. Research–policy dominance $\leftrightarrow$ Semantic gap			
MPNet LR	+0.274	+0.196	+0.115
MPNet MLP	+0.050	+0.090	-0.014
MPNet ZS	-0.152	-0.227	+0.033
MiniLM LR	+0.293	+0.119	+0.240
MiniLM MLP	+0.353	+0.150	+0.260
SciBERT LR	+0.460†	-0.156	+0.420†
SciBERT MLP	+0.194	-0.358	+0.484*
Concept LR	+0.117	+0.120	+0.021
Concept MLP	-0.060	+0.046	-0.119
H1c. Research coverage $\leftrightarrow$ Semantic gap			
MPNet LR	+0.288	+0.464†	-0.055
MPNet MLP	+0.032	+0.308	-0.197
MPNet ZS	-0.191	-0.117	-0.087
MiniLM LR	+0.302	+0.355	+0.101
MiniLM MLP	+0.429†	+0.438†	+0.132
SciBERT LR	+0.419†	+0.208	+0.029
SciBERT MLP	+0.029	-0.234	+0.262
Concept LR	+0.134	+0.262	-0.085
Concept MLP	-0.074	+0.113	-0.205
H1d. Policy coverage $\leftrightarrow$ Semantic gap			
MPNet LR	-0.018	+0.501*	-0.359
MPNet MLP	-0.045	+0.418†	-0.361
MPNet ZS	-0.052	+0.258	-0.244
MiniLM LR	-0.041	+0.381	-0.284
MiniLM MLP	+0.064	+0.466*	-0.269
SciBERT LR	-0.138	+0.728**	-0.813***
SciBERT MLP	-0.344	+0.287	-0.492*
Concept LR	+0.009	+0.269	-0.225
Concept MLP	-0.017	+0.130	-0.154

Notes: Same H1a–H1d grid and column layout as Table 4; significance stars use the two-sided Monte Carlo permutation p-value ($^{***}\,p < 0.001$, $^{**}\,p < 0.01$, $^{*}\,p < 0.05$, $^{\dagger}\,p < 0.10$). Identical inputs and configuration grid to Table 4; only the correlation function changes from Spearman ρ (ranks) to Pearson r (raw magnitudes). ZS appears here for MPNet only. Appendix I pools it across encoders.

Table 27. Per-SDG policy source-family sensitivity: coverage share and semantic gap.

SDG	Full Corpus		Curated AI/SDG		SDGi VNR/VLR		UNGDC	
	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)	pol.% (n)	sem. (n)
1	4.5 (288)	0.415 (1,689)	1.1 (1)	0.443 (357)	5.9 (250)	0.415 (1,191)	1.8 (37)	0.556 (141)
2	4.3 (274)	0.299 (1,508)	0.0 (0)	0.345 (240)	5.8 (246)	0.294 (1,196)	1.4 (28)	0.454 (72)
3	4.7 (298)	0.495 (2,408)	6.4 (6)	0.362 (643)	6.7 (284)	0.554 (1,707)	0.4 (8)	0.695 (58)
4	4.8 (305)	0.267 (2,055)	0.0 (0)	0.205 (274)	7.0 (296)	0.287 (1,718)	0.4 (9)	0.395 (63)
5	5.1 (324)	0.278 (2,005)	0.0 (0)	0.276 (272)	6.6 (278)	0.282 (1,531)	2.2 (46)	0.411 (202)
6	3.7 (238)	0.292 (1,206)	0.0 (0)	0.346 (137)	5.3 (226)	0.292 (1,038)	0.6 (12)	0.422 (31)
7	4.3 (273)	0.427 (1,578)	1.1 (1)	0.511 (358)	6.0 (253)	0.409 (1,147)	0.9 (19)	0.591 (73)
8	4.3 (276)	0.387 (1,886)	0.0 (0)	0.395 (309)	6.5 (275)	0.397 (1,558)	0.0 (1)	0.536 (19)
9	4.3 (274)	0.315 (2,986)	50.0 (47)	0.272 (1,838)	5.1 (215)	0.545 (1,116)	0.6 (12)	0.459 (32)
10	2.9 (182)	0.272 (1,085)	0.0 (0)	0.331 (258)	4.2 (176)	0.278 (774)	0.3 (6)	0.402 (53)
11	5.3 (338)	0.449 (1,658)	0.0 (0)	0.428 (175)	8.0 (336)	0.456 (1,474)	0.1 (2)	0.573 (9)
12	3.5 (226)	0.296 (1,132)	0.0 (0)	0.384 (178)	5.3 (225)	0.297 (947)	0.0 (1)	0.488 (7)
13	11.3 (718)	0.480 (3,084)	7.4 (7)	0.437 (613)	6.0 (252)	0.444 (1,268)	22.4 (459)	0.576 (1,203)
14	3.8 (241)	0.360 (1,293)	0.0 (0)	0.372 (216)	4.5 (189)	0.354 (856)	2.5 (52)	0.487 (221)
15	4.2 (266)	0.238 (1,484)	0.0 (0)	0.289 (299)	5.7 (239)	0.237 (1,108)	1.3 (27)	0.409 (77)
16	17.3 (1,100)	0.329 (4,890)	12.8 (12)	0.195 (1,092)	5.9 (250)	0.377 (1,395)	40.9 (838)	0.506 (2,403)
17	11.7 (746)	0.177 (3,645)	21.3 (20)	0.181 (859)	5.6 (235)	0.165 (1,322)	24.0 (491)	0.260 (1,464)
ρ	1.000	1.000	0.355	0.733	0.613	0.914	0.453	0.966

Notes: For each policy source family, `pol.% (n)` is the document-weighted policy share (%) with the document count in parentheses; `sem. (n)` is the within-SDG research–policy semantic gap (naive, unadjusted) with the capped segment count in parentheses. Read across a row to see whether the gap is stable or driven by one family.

Table 28. Per-family interaction test: coverage predictors vs the within-SDG semantic gap, raw and register-adjusted.

Family	n	Research share (ρ, p)	Coverage gap (ρ, p)	Coverage gap adj. (ρ, p)	Research share excl. SDG 4 (ρ, p)
Full corpus	17	0.449 (0.071)	-0.012 (0.959)	0.544 (0.025)	0.606 (0.015)
Curated AI/SDG	17	-0.056 (0.833)	-0.223 (0.386)	-0.061 (0.807)	0.053 (0.848)
SDGi VNR/VLR	17	0.596 (0.014)	0.042 (0.873)	0.498 (0.044)	0.703 (0.003)
UNGDC speeches	15	0.425 (0.115)	0.354 (0.195)	0.357 (0.191)	0.609 (0.024)

Notes: Spearman ρ (with two-sided Monte Carlo permutation p, 100,000 resamples, fixed seed) between each coverage predictor and the within-SDG semantic gap across the reliable SDGs (n). Research share is the research-corpus proportion per SDG (family-independent); the coverage gap is $|\text{research} - \text{policy}|$ (percentage points) using the family's document-weighted policy share. The Coverage gap adj. column correlates that same coverage gap with the register-adjusted gap obtained by projecting both sides through the INLP matrix (Section 3.8); raw and adjusted rows use identical capped segment sets. The curated AI/SDG family is the symmetric AI/SDG-vs-AI/SDG control matching the research corpus's AI $\cap$ SDG scope.

I POOLED REGRESSION: COVERAGE PREDICTORS AND THE SEMANTIC GAP

This appendix pools all 24 encoder–retrieval–method–cap configurations into a single panel ($N = 408$) and regresses the within-SDG semantic gap on coverage predictors using OLS with cluster-robust standard errors by SDG (17 clusters). The method dimension includes the zero-shot nearest-centroid rule under each of the three encoders, making this appendix the one place the method spans encoders. The dependent variable is the rank of the semantic gap within each configuration (1–17 per config, ties averaged); predictors are expressed in percentage points (0–100). Table 29 reports the full 23-specification grid across core regressions (Panel A), robustness checks (Panel B), interaction terms (Panel C), and alternative functional forms (Panel D).

The adjusted semantic gap is positively associated with the coverage gap ($b = +0.18$, $p = 0.069$; Panel A, column 1). The coefficient is marginally significant at conventional thresholds but is confirmed by the supervised-only subsample ($p = 0.040$), the MPNet-only subsample ($p = 0.003$), and the raw-DV specification in cosine units ($p = 0.007$). The raw gap and the register component show no detectable association with any coverage predictor, consistent with the raw-null pattern reported in Section 4.4. Policy coverage (the share of policy documents assigned to a given SDG) is a stronger predictor of the adjusted gap ($b = +0.41$, $p = 0.003$), capturing how concentrated policy attention is on each goal. The coverage-gap-by-encoder interaction (Panel C) shows that the effect is strongest for MPNet and reverses for SciBERT, suggesting that domain-specialised pretraining compresses the adjusted semantic gap regardless of coverage structure. This reversal is a statement about interaction coefficients (each encoder's slope relative to the pooled average), not about the per-configuration Spearman signs in Table 4, which stay positive for SciBERT.

Table 29. Pooled OLS: coverage predictors and the within-SDG semantic gap rank across 24 configurations.

	Core									Robustness						Interactions				Functional form			
	(1) Adj. gap covgap	(2) Adj. gap dominance	(3) Raw gap covgap	(4) Raw gap dominance	(5) Reg. gap covgap	(6) Reg. gap dominance	(7) Adj. gap rescov	(8) Raw gap rescov	(9) Reg. gap rescov	(10) No classifier ind.	(11) + SDG FE	(12) Supervised only	(13) Keyword only	(14) MPNet only	(15) No clf + SDG FE	(16) + covgap×enc	(17) + covgap×ret	(18) + covgap×mth	(19) + all interactions	(20) Raw DV (not ranked)	(21) Log DV	(22) WLS	(23) Rank + bootstrap
Coverage gap (|research% − policy%|)	+0.185†	–	+0.093	–	-0.044	–	–	–	–	+0.185†	-0.010	+0.199*	+0.167	+0.329**	-0.010	+0.333**	+0.170	+0.239**	+0.419***	+0.003**	+0.003**	+0.389***	+0.185†
	(0.095)		(0.169)		(0.112)					(0.095)	(0.188)	(0.089)	(0.102)	(0.096)	(0.188)	(0.099)	(0.101)	(0.081)	(0.092)	(0.001)	(0.001)	(0.081)	(0.095)
Policy coverage	+0.406**	+0.544**	-0.090	+0.065	-0.313†	-0.295	+0.464**	-0.076	-0.336†	+0.406**	+1.210†	+0.393**	+0.427**	+0.428*	+1.210†	+0.399**	+0.406**	+0.406**	+0.399**	+0.004*	+0.003*	+0.478***	+0.406**
	(0.116)	(0.177)	(0.182)	(0.163)	(0.179)	(0.182)	(0.138)	(0.173)	(0.179)	(0.116)	(0.656)	(0.114)	(0.118)	(0.150)	(0.656)	(0.109)	(0.116)	(0.116)	(0.108)	(0.002)	(0.001)	(0.087)	(0.116)
MiniLM (vs. MPNet)	+0.070	-0.000	+0.035	-0.000	-0.017	+0.000	-0.000	-0.000	+0.000	+0.070	-0.004	+0.075	+0.063	–	-0.004	+0.361	+0.064	+0.067	+0.531	-0.035**	-0.029**	+0.237	+0.070
	(0.798)	(0.777)	(1.155)	(1.133)	(0.922)	(0.917)	(0.777)	(1.133)	(0.917)	(0.798)	(0.781)	(0.896)	(0.794)		(0.781)	(1.118)	(0.795)	(0.798)	(1.086)	(0.011)	(0.009)	(0.904)	(0.798)
SciBERT (vs. MPNet)	-0.097	-0.000	-0.049	-0.000	+0.023	+0.000	-0.000	-0.000	+0.000	-0.097	+0.005	-0.104	-0.087	–	+0.005	+2.596*	-0.089	-0.092	+2.766*	-0.146***	-0.128***	-3.409	-0.097
	(1.146)	(1.139)	(0.999)	(0.979)	(1.288)	(1.294)	(1.139)	(0.979)	(1.294)	(1.146)	(1.131)	(1.261)	(1.147)		(1.131)	(1.196)	(1.147)	(1.146)	(1.188)	(0.016)	(0.013)	(2.083)	(1.146)
Concept (vs. keyword)	-0.034	+0.000	-0.017	-0.000	+0.008	-0.000	+0.000	+0.000	-0.000	-0.034	+0.002	-0.036	–	-0.060	+0.002	-0.061	-0.627	-0.040	+0.764	+0.107***	+0.085***	-1.020	-0.034
	(0.669)	(0.662)	(0.777)	(0.800)	(0.952)	(0.960)	(0.662)	(0.800)	(0.960)	(0.632)	(0.688)	(0.678)		(0.755)	(0.660)	(0.681)	(1.019)	(0.674)	(0.802)	(0.014)	(0.010)	(1.217)	(0.669)
Cap 20 (vs. 50)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	+0.000	-0.000	+0.000	-0.000	+0.000	-0.000	+0.000	-0.000	-0.000	-0.000	+0.003*	+0.003*	-0.055	-0.000
	(0.102)	(0.102)	(0.100)	(0.100)	(0.160)	(0.160)	(0.102)	(0.100)	(0.160)	(0.184)	(0.102)	(0.102)	(0.109)	(0.137)	(0.184)	(0.102)	(0.102)	(0.102)	(0.102)	(0.001)	(0.001)	(0.332)	(0.102)
No cap (vs. 50)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	+0.000	+0.000	-0.000	+0.000	-0.000	+0.000	-0.000	-0.000	+0.000	-0.000	+0.000	+0.001	+0.001	-0.157	-0.000
	(0.248)	(0.248)	(0.193)	(0.193)	(0.151)	(0.151)	(0.248)	(0.193)	(0.151)	(0.279)	(0.248)	(0.248)	(0.277)	(0.232)	(0.279)	(0.248)	(0.248)	(0.248)	(0.248)	(0.003)	(0.002)	(0.406)	(0.248)
MLP (vs. LR)	-0.000	-0.000	-0.000	+0.000	-0.000	-0.000	+0.000	+0.000	+0.000	–	-0.000	+0.000	-0.000	+0.000	–	-0.000	-0.000	+0.569	+0.494	+0.012*	+0.010*	-0.821	-0.000
	(0.395)	(0.395)	(0.479)	(0.479)	(0.309)	(0.309)	(0.395)	(0.479)	(0.309)		(0.395)	(0.395)	(0.395)	(0.564)		(0.395)	(0.395)	(0.512)	(0.537)	(0.004)	(0.004)	(0.878)	(0.395)
ZS (vs. LR)	+0.000	+0.000	+0.000	-0.000	+0.000	-0.000	+0.000	+0.000	-0.000	–	-0.000	–	+0.000	+0.000	–	-0.000	-0.000	+0.778	+0.703	+0.003	+0.003	-0.787	+0.000
	(0.723)	(0.723)	(1.302)	(1.302)	(0.823)	(0.823)	(0.723)	(1.302)	(0.823)		(0.723)		(0.729)	(0.980)		(0.723)	(0.723)	(0.896)	(0.910)	(0.008)	(0.006)	(1.283)	(0.723)
Domination (research% − policy%)	–	+0.080	–	+0.141	–	+0.041	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
		(0.097)		(0.104)		(0.063)																	
Research coverage (%)	–	–	–	–	–	–	+0.080	+0.141	+0.041	–	–	–	–	–	–	–	–	–	–	–	–	–	–
							(0.097)	(0.104)	(0.063)														
sdg1	–	–	–	–	–	–	–	–	–	–	+7.546†	–	–	–	+7.546†	–	–	–	–	–	–	–	–
											(3.860)				(3.860)								
sdg2	–	–	–	–	–	–	–	–	–	–	-2.821	–	–	–	-2.821	–	–	–	–	–	–	–	–
											(4.283)				(4.283)								
sdg3	–	–	–	–	–	–	–	–	–	–	+7.301	–	–	–	+7.301	–	–	–	–	–	–	–	–
											(4.472)				(4.472)								
sdg4	–	–	–	–	–	–	–	–	–	–	+1.540	–	–	–	+1.540	–	–	–	–	–	–	–	–
											(3.615)				(3.615)								
sdg5	–	–	–	–	–	–	–	–	–	–	+4.778	–	–	–	+4.778	–	–	–	–	–	–	–	–
											(3.667)				(3.667)								
sdg6	–	–	–	–	–	–	–	–	–	–	-0.994	–	–	–	-0.994	–	–	–	–	–	–	–	–
											(4.599)				(4.599)								
sdg7	–	–	–	–	–	–	–	–	–	–	+1.634	–	–	–	+1.634	–	–	–	–	–	–	–	–
											(4.377)				(4.377)								
sdg8	–	–	–	–	–	–	–	–	–	–	+2.005	–	–	–	+2.005	–	–	–	–	–	–	–	–
											(4.178)				(4.178)								
sdg9	–	–	–	–	–	–	–	–	–	–	+1.453	–	–	–	+1.453	–	–	–	–	–	–	–	–
											(4.148)				(4.148)								
sdg10	–	–	–	–	–	–	–	–	–	–	+10.103†	–	–	–	+10.103†	–	–	–	–	–	–	–	–
											(5.245)				(5.245)								
sdg11	–	–	–	–	–	–	–	–	–	–	+1.950	–	–	–	+1.950	–	–	–	–	–	–	–	–
											(3.570)				(3.570)								
sdg12	–	–	–	–	–	–	–	–	–	–	+3.523	–	–	–	+3.523	–	–	–	–	–	–	–	–
											(4.729)				(4.729)								
sdg13	–	–	–	–	–	–	–	–	–	–	-1.594†	–	–	–	-1.594†	–	–	–	–	–	–	–	–
											(0.760)				(0.760)								
sdg14	–	–	–	–	–	–	–	–	–	–	+0.809	–	–	–	+0.809	–	–	–	–	–	–	–	–
											(4.588)				(4.588)								
sdg15	–	–	–	–	–	–	–	–	–	–	-2.840	–	–	–	-2.840	–	–	–	–	–	–	–	–
											(4.395)				(4.395)								
sdg16	–	–	–	–	–	–	–	–	–	–	-9.426	–	–	–	-9.426	–	–	–	–	–	–	–	–
											(5.690)				(5.690)								
covgap $\times i_{\text{minilm}}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.045	–	–	-0.075	–	–	–	–
																(0.091)			(0.096)				
covgap $\times i_{\text{scibert}}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.452***	–	–	-0.482***	–	–	–	–
																(0.095)			(0.102)				
covgap $\times i_{\text{concept}}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	+0.103	–	-0.145†	–	–	–	–
																	(0.076)		(0.075)				
covgap $\times i_{\text{mlp}}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.101†	-0.087†	–	–	–	–
																		(0.048)	(0.047)				
covgap $\times i_{\text{zs}}$	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	-0.138	-0.124	–	–	–	–
																		(0.104)	(0.103)				
N	408	408	408	408	408	408	408	408	408	408	408	357	357	170	408	408	408	408	408	408	408	408	408
R^2	0.195	0.162	0.014	0.049	0.073	0.074	0.162	0.049	0.074	0.195	0.617	0.198	0.196	0.331	0.617	0.255	0.197	0.200	0.260	0.684	0.701	0.521	0.195
Adj. R^2	0.177	0.143	-0.008	0.028	0.052	0.053	0.143	0.028	0.053	0.181	0.592	0.179	0.177	0.302	0.594	0.235	0.177	0.178	0.233	0.677	0.694	0.510	0.177

J DECLARATION OF AI USE

This declaration is made in accordance with the University of Birmingham *Code of Practice on Academic Integrity 2025–26* (Appendix A). AI tools were used only as aids to the author's own work: every idea, argument, research question, analysis design, result, and interpretation in this dissertation is the author's own. AI-generated suggestions and code were reviewed, tested, and independently verified by the author before inclusion, and no AI-generated text or analysis was submitted as the author's work without the author's full authorship and oversight. This declaration lists the ways AI tools were used. Uses not listed (e.g., code debugging, plotting, LaTeX troubleshooting, literature summarisation) were subject to the same oversight and verification.

Table 30. Declaration of AI use.

Activity	AI tool(s)	How output was used
Researching and refining ideas	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Used to explore and stress-test ideas and to check data/code availability for feasibility. All adopted ideas, arguments, and conclusions are the author's own and were independently evaluated; the final research design was presented to and approved by the supervisor.
Information retrieval / background research	Gemini, MiniMax / DeepSeek (via OpenCode), GPT (via Codex), Hy3 (via OpenCode), Claude (Desktop)	AI tools read the manuscript for context and recommended papers to investigate further. All cited sources were independently verified against the original publications.
Drafting an outline / organising thoughts	MiniMax / DeepSeek (via OpenCode), GPT (via Codex), Hy3 (via OpenCode), Mimo (via OpenCode), GLM (via OpenCode)	AI coding agents generated code scaffolding that the author reviewed, unit-tested, and directed; all analytical decisions, results, and interpretation are the author's own.
Refining research questions	ChatGPT, MiniMax / DeepSeek (via OpenCode), GPT (via Codex)	Used to explore alternative framings and to confirm the questions were answerable with the available data and models. The author retained full control over the final formulation, and the research questions were presented to and approved by the supervisor.
Checking spelling and grammar	ChatGPT, Gemini, Claude (web/mobile), MiniMax / DeepSeek (via OpenCode), GPT (via Codex), Hy3 (via OpenCode), Mimo (via OpenCode), GLM (via OpenCode)	Used for spelling, punctuation, grammar, and syntax checking, and for formatting/ordering of references and cross-references. All suggestions were accepted or rejected by the author; AI was not used to alter or develop the ideas or arguments presented.